

The Mathematics of Tatiana

A Cellular–Sheaf Cognitive Architecture over a
Stratified Simplicial Complex

*A complete, implementation-free consolidation,
with epistemic status attached to every claim*

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How to read this book

What this is

This book consolidates the entire mathematical content of TATIANA into a single document. Its sources are the files in `DOCS/`, the poster sections in `POSTER/`, and the specification documents added on the branch `fix-11-13-and-curvature-decisions`. It is *implementation-free*: no C++, no Python. Where the implementation determines a mathematical fact — and it sometimes does — the fact is stated abstractly here and the discrepancy is recorded in Appendix B.

Why every claim carries a tag

The source documents *contradict one another*. The internal audit (`DOCS/AUDIT_SCRUTINY_AND_BOOK.md`) records fourteen findings against the earlier mathematical description, several of which are outright refutations: a proposition false in the deployed case, a theorem stated more generally than it is proved, a repair for $H^1(\mathcal{C}; \mathcal{F})$ that is half circular. A smooth merge of these sources would present refuted material in the same confident register as proved material, and a reader studying it would learn falsehoods.

So every substantive claim in this book carries one of six tags:

[PROVED]	A complete proof is given here, from stated hypotheses.
[PROVED UNDER STATED CONDITIONS]	Proved, but only under hypotheses the deployed system may not satisfy.
[IMPLEMENTED, UNPROVED]	The system computes this; no proof of correctness exists.
[PROPOSED]	A design intention. Not built, not proved.
[CONTESTED BY AUDIT]	The audit disputes this. The dispute is stated in place.
[REFUTED]	Tested and failed, or shown false. Retained <i>with</i> the refutation.

Refuted material is kept rather than deleted. Knowing why a construction failed is most of what you need in order not to rebuild it.

The three layers

Each central object is presented three times over, so that a single reading can be shallow or deep:

1. a boxed **Intuition** in plain language, with no symbols;
2. the formal **Definition**, **Construction** or **Proposition**, with proof;

3. a **Worked example** on an explicit small complex, carried through to numbers.

The worked examples are cumulative. A single running example — a seven-organ complex called K_7 , introduced in Chapter 2 — is reused throughout, so that the same object is measured by every instrument the book builds. Where an exercise appears, it is because the step is short and doing it yourself is faster than reading it.

Provenance

Grey boxes marked **Source** record which file a passage descends from, and what was changed in merging it. They exist so that any statement here can be traced back and checked against the original, and so that when the sources disagree you can see *that* they disagree rather than only seeing the winner.

This book is mid-rewrite, and says so on every page

The book was written for **V0**: an architecture in which typed organs were dispatched by external code around a language model. That architecture is retired. Chapter 1 states what replaced it and why.

Most of the mathematics survives the pivot. The framing around it does not, and in several places the *justification* for a construction was organ-shaped even where the construction itself was not. So every chapter now opens with two things:

- a **disposition** — **iKEEP_i**, **iRECAST_i**, **iRETIRE_i** or **iNEW IN V2_i** — saying what the pivot does to it;
- a **work plan** box: the goal, the raw material, the ordered steps, and the condition under which the chapter is finished.

A work plan is an *instruction for writing the chapter*, not commentary on it. When a chapter is written, its work plan is deleted and replaced by the prose it asked for. **A chapter that still contains a work plan is a chapter that is not done.** This is deliberate: it makes the book's own incompleteness impossible to overlook, including by its author.

Reading order

Chapter 1 first, always — it is short and everything else is answerable to it.

Then Parts II and III (state space, coherence) as the conceptual spine, in order. Part V (cohomology) carries Proposition 8.2, which is the constraint the entire architecture is built around, and should be read before anything that mentions growth.

Parts VI and VII — **Identity** and **The Two-Phase Law** — are the centre of V2. They are new, and a reader who reads only three parts should read Chapter 1 and those two.

The remaining parts may be read in any order once the spine is in place, with one exception: the limits chapter is the honest boundary and should be read before quoting any result from this book to anyone else.

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Part I

What TATIANA Is

Chapter 1

The V2 Reformulation

1.1 What changed, and why the book must change with it

This book was written for **V0**: an architecture in which a C++ dispatcher called out to typed *organs* — Search, Reason, Context, Verify — one of which was a remote language model. Under that framing the mathematics of this book was a *library*: the dispatcher decided when to run the growth law, when to build a cone, when to fire the adjunction. The sheaf was something the code consulted.

V0 is retired. The reason is not that its mathematics failed — most of it survives, and Part 21 records exactly which parts did fail — but that the architecture was the wrong shape for the thing it was built to be. What was built was a complex of organs wrapped around a borrowed brain. What is wanted is a single kernel that, meeting a problem, mutates, grows and reproduces itself to cover it, while remaining provably the same kernel throughout.

Intuition. The difference is who is in charge. In V0, the mathematics was a set of tools and a program decided when to pick up each one. In V2, there is no program outside the mathematics: the kernel’s identity *is* “the thing that keeps applying its own laws to itself.” Nothing conducts it.

Source. DOCS/TATIANA.LOGBOOK.md §8 records the decision; DOCS/TATIANA.V1.IDENTITY_MAP.md is the binding statement of what V2 is and is the document this book is now being rewritten to match; DOCS/MATH.TOOLBOX.md is the complete inventory of what V0 proved, measured and killed, and is the source of every “already established” citation below. The logbook calls the successor V1 and the working folder calls it V2. They are the same pivot. This book says **V2** throughout.

1.2 The commitment

Everything in this book is now answerable to one sentence.

TATIANA is a cellular sheaf on a stratified 2-complex whose section relaxes, whose restriction maps mold, and whose complex grows and prunes — under a two-phase law that conserves a stated invariant exactly while awake and projectively while offline.

Read the clauses as constraints. Each one forbids something, and each forbidding is a live constraint on the chapters that follow.

clause	forbids
cellular sheaf on a 2-complex section relaxes	dropping to a graph treating s as stored data rather than a dynamical variable
restriction maps mold complex grows and prunes	freezing $\mathcal{F}$, which makes the sheaf decorative growth-only architectures that inflate without bound
two-phase stated invariant	trading all costs against each other on every tick “identity” as a vibe rather than a number that can be printed

1.3 What tatiana is not

Each of these was a live position in this project at some point, which is why each is recorded rather than assumed.

1. **Not a language-model wrapper.** No router, no retrieval-augmented arm, no “memory layer for a chat model.” The language model is never the thing that reasons. Two roles remain structurally necessary — an *encoder* producing the observation o_t , and a *verification lift* — and V2’s position on them is given in §1.5: they are required, they are not to be filled by plugging in a pretrained model, and their own architecture is an open research question.
2. **Not a knowledge graph.** A knowledge graph has no dynamics and no obstruction theory. If a proposed feature would work identically on a labelled graph, it does not belong here.
3. **Not a cognitive-science model.** Mechanisms are borrowed from neuroscience as design constraints. No claim is made to explain brains, and nothing here should be evaluated against neural data.
4. **Not the Free Energy Principle.** The two-phase objective contains structural and stability terms that are this project’s construction. Calling it free energy imports authority that has not been earned. It is called the *maintenance cost* and nothing else.
5. **Not a system that must be conscious, agentic or “alive.”** Reproduction is a morphism. No further reading is claimed.

1.4 The six structural changes

These are the changes that force a rewrite rather than an edit.

1. **One kernel, not a complex of organs.** The coarse/fine stratification in Chapter 2 was justified by organs being real, separate subsystems. With one kernel that justification is gone. Whether the stratification survives at all — and if so, on what new ground — is a question Part 2 must now answer rather than assume.

2. **Identity is a number, and it is the centre of the book.** V0 had no invariant. V2’s is

$$K(\mathbb{K}) = (\dim \mathcal{F}_{\mathbb{K}}(\sigma))_{\sigma \in \mathcal{C}_{\mathbb{K}}},$$

the dimension vector over the crystallized store, exactly conserved by molding and consolidation and changed only by growth, and then by a derived amount. Identity is carried by $\mathbb{K}$; the working complex W may be destroyed at the end of every session. This has no chapter in the V0 book and needs one at the front, not the back.

3. The dynamics are two-phase, not a single objective. TATIANA does not minimise one functional every tick. It alternates: a **wake** phase that discharges discord by the Hodge routing law, and a **sleep** phase that renormalizes and prunes. Cost lives in sleep and nowhere else. Growth adds during wake, pruning removes during sleep, and they never bid against each other inside one tick.

Claims and non-claims. A six-term λ -weighted energy functional was proposed for this role and **refused**. The refusal is load-bearing, not stylistic: a second exchange rate is evidence that the two-phase separation has failed. Chapter 9 states the one exchange rate that is allowed and Part 21 records why the alternative was rejected. Do not reintroduce it.

4. Growth has an address problem, and it is now load-bearing. Proposition 8.2 — $[\delta x] = 0$ in $H^1(\mathcal{C}; \mathcal{F})$ unconditionally, so a coboundary can never name a gluing failure — was, in V0, one blocked mechanism among several. With no organ complex to lean on, a kernel that cannot address its own growth has no growth at all. The fix requires a genuinely *measured* 1-cochain that is not the coboundary of any global state, and the harmonic component of its Hodge decomposition is the address. Under V0 that cochain came from organs disagreeing with each other. V2 has no organs, so where it comes from is open.

5. The 2-cells are load-bearing, not decoration. On a graph, $\delta^1 \delta^0 = 0$ is vacuous: every assignment of stalks and maps is a sheaf, molding is unconstrained, and nothing intrinsic can ever stall. On a 2-complex it is a real algebraic condition. The triangles are what make molding *obstructible*, hence what make growth *derivable*. Every published negative result about learned sheaf Laplacians concerns graphs, and this is the defence — which means it must be stated as a theorem and not as a hope.

6. Reproduction enters as a morphism. The adjoint pair already in the mathematics is the candidate: ι^* instantiates a subcomplex, $\iota_!$ writes back only what was touched, and $\iota^* \iota_! \cong \text{Id}$ guarantees the round trip does not corrupt the fragment. An offspring is a downward-closed subcomplex with its pullback sheaf, carrying a sub-invariant. No claim is made that it is viable. This is the weakest part of the design and is placed last for that reason.

1.5 Where the language model is, and is not

Claims and non-claims. The reasoner is never a language model. That is settled and is not reopened anywhere in this book.

Two roles remain, and pretending otherwise would be dishonest: something must turn raw input into an observation o_t that the sheaf can act on (an *encoder*), and something must discharge the verification trichotomy (a *lift*). V2's position is narrower than V0's and is

stated here so it cannot drift: both roles are required; neither is to be satisfied by wrapping a pretrained, externally-hosted model; and the architecture of each is an *open research question* to be derived from this project's own mathematics the way every other component was. It is possible that what is derived ends up structurally resembling machinery that also appears in language models. If so, that must be an earned conclusion with a stated derivation — never a default, and never an import.

1.6 How the rewrite proceeds

Every chapter that follows carries a **work plan** box and a **disposition**:

KEEP	The mathematics stands. Reframing only, if anything.
RECAST	The result survives but its justification or its role changes.
RETIRE	Belongs to V0's architecture and does not survive the pivot.
NEW IN V2	No V0 counterpart. Must be written from scratch.

The work plan states the goal, the raw material, the ordered steps, and the condition under which the section is finished. It is an instruction, not commentary: when a section is written, its work plan is deleted and replaced by the prose it asked for. A chapter with a work plan still in it is a chapter that is not done.

Claims and non-claims. Two rules govern the order of work, both earned the hard way in V0 and both recorded in MATH_TOOLBOX.md Part IX.

Write the falsifier before the implementation. Every construction that claims to be measurable states, before it is built, what result would refute it. Building an instrument and a result at the same time is how V0 produced four confident wrong answers in one week.

New mathematics earns its place only if it changes a number a program prints, or deletes a decision a human was making by hand. A construction that does neither is decoration, however correct.

Part II

The State Space

Chapter 2

The Base Complex

Work plan. ¡RECAST!

Goal. Rebuild the base complex on a justification that survives the loss of organs. The mathematics of $\mathcal{C}$ is sound; the story told about *why* it is stratified is V0's and is gone.

Raw material. This chapter's existing text (the simplicial-complex material and the running example are reusable as-is); MATH_TOOLBOX.md I.1 (simplices as honest n -ary relations — the one place the topology is earned rather than borrowed; also the list of what was cut as decorative); TATIANA_V1_IDENTITY_MAP.md §4.3 (2-cells are load-bearing).

Steps.

1. **Replace the opening intuition.** The current one describes “a small society of specialists” — search, memory, reason, check. That is V0's organ complex and it is retired. Write a new intuition for one kernel whose parts are not specialists but regions of a single body of thought.
2. **Keep, unchanged, the n -ary binding argument.** Simplices are honest joint relations, not decomposed pairwise facts: a theorem depending on three prerequisites is one 3-simplex, not three edges. This is the strongest justification in the chapter and it does not depend on organs at all. ¡KEEP!
3. **Re-derive or retire the stratification.** §2.3 currently justifies coarse/fine by organs being real separate subsystems. With one kernel that ground is gone. Decide, and write down the decision with its reason: either the stratification survives on a new ground (scale of abstraction? crystallized vs working, which is a *different* axis and was repeatedly confused with this one in V0), or it collapses to a single scale. Do not leave this ambiguous — V0 conflated the two axes repeatedly and had to keep re-separating them.
4. **Promote the 2-cells.** The current text treats the complex as up-to-2-dimensional almost incidentally. In V2 the 2-cells are what make molding obstructible, hence what make growth derivable, hence the defence against every published negative result about learned sheaf Laplacians on graphs. State that here, forward-referencing Chapter 9, so the reader knows from the start why triangles are not optional.
5. **Keep the running example K_7 ,** but re-describe its vertices as something other than organs. Every later chapter measures this object; changing it is expensive, so change only its interpretation.
6. **Carry forward the register of what was cut and why.** Infinite-dimensional Banach stalks, Pachner moves (a knowledge graph is not a manifold, so the preservation guarantee does not transfer), cohomology on undefined restriction maps. Keep these as refutations, not silence.

Done when. The chapter justifies every structural feature of $\mathcal{C}$ without once appealing to organs, and the stratification question is answered rather than inherited.

2.1 What the object is

Intuition. Picture a research assistant that is not one mind but a small society of specialists: one that searches, one that remembers, one that reasons, one that checks. TATIANA is the mathematics of *that society* — who is bound to whom, whether their beliefs agree, by how much they disagree and exactly where, and how the society rewires itself as it works. The reasoning itself is done by an external language model. The mathematics governs the society, not the neurons.

Definition 2.1 (Cognitive state space). A *cognitive state space* is a triple $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$ where

- (i) $\mathcal{C}$ is a stratified finite abstract simplicial complex (Section 2.3);
- (ii) $\mathcal{F}$ is a cellular sheaf of finite-dimensional real inner-product spaces on $\mathcal{C}$ (Chapter 3);
- (iii) $s \in C^0(\mathcal{C}; \mathcal{F})$ is a distinguished 0-cochain, the *current state*, whose coherence is measured by the sheaf Laplacian (Chapter 4).

The space evolves in discrete time under structural operators drawn from an operator algebra $\mathfrak{D}$ which is itself permitted to grow (Part V).

Source. DOCS/TATIANA.Mathematical.Description.tex, Def. 1.1, merged with DOCS/TATIANA_LOGBOOK.md §0.1. The logbook writes the triple as (C, F, s) with C “a stratified simplicial complex (coarse level = cognitive organs as vertices; fine level = concepts inside each organ’s stalk)”. Note that this phrasing conflates two different things — the fine complex $\mathcal{C}_v$ and the stalk $\mathcal{F}(v)$ — which are kept strictly separate here and in Chapter 3. See Appendix B, item D1.

2.2 Simplicial complexes and n -ary binding

Intuition. A graph records *pairwise* facts: A relates to B . But a theorem depending on three hypotheses is *one* three-way fact, not three separate pairs — drop any one hypothesis and the fact is gone, which is not how three independent edges behave. A simplicial complex is the smallest honest language for facts of arbitrary arity: a filled triangle is a genuine three-way binding.

Definition 2.2 (Abstract simplicial complex). Let V be a finite set. An *abstract simplicial complex* $\mathcal{C}$ on V is a family of nonempty subsets of V , called *simplices*, which is *downward closed*: if $\sigma \in \mathcal{C}$ and $\emptyset \neq \tau \subseteq \sigma$, then $\tau \in \mathcal{C}$. A simplex of cardinality $k + 1$ is a k -*simplex*; write $\dim \sigma = k$. If $\tau \subseteq \sigma$ we write $\tau \trianglelefteq \sigma$ and call τ a *face* of σ . The set of k -simplices is denoted $\mathcal{C}_k$.

Thus 0-simplices are vertices, 1-simplices edges, 2-simplices filled triangles, and so on. Downward closure is not a technical convenience; it is the modelling assumption that every sub-collection of a jointly bound fact is itself recorded.

Remark 2.3 (Simplices are n -ary relations). A k -simplex $\{v_0, \dots, v_k\}$ models a single $(k+1)$ -ary binding among its vertices, not $\binom{k+1}{2}$ pairwise ones. This distinction is what later earns the use of b_1 as a diagnostic (Part IV): a 1-cycle counts as a genuine structural hole only because a *filled* 2-simplex means “these three are jointly bound” rather than merely “each pair is bound”.

Remark 2.4 (Two relations on the same vertices). A logical prerequisite chain is *directed*: A is needed for B . A simplicial complex is undirected. TATIANA therefore carries two relations on

the same vertex set — a directed poset for logical dependency, and the undirected simplices for n -ary co-binding. They are never conflated, and no result in this book uses one where the other is meant.

2.3 The two-level stratification

Intuition. The society has two scales. Zoom out and the vertices are whole *organs*. Zoom into one organ and it is itself a little complex whose vertices are *concepts*. The same geometry runs at both scales, and the two are glued by a coarse-graining map developed in Part III.

Construction 2.5 (The stratified complex). $\mathcal{C}$ carries two strata.

- The *coarse complex* K has as vertices the cognitive organs

$$V(K) = \{\text{PLANNER, SEARCH, CANONICALIZER, CURATOR, WORKINGMEMORY, REASONING, VERIFY}\}.$$

A simplex $\sigma \in K$ is a *bound coalition* of organs acting as a unit, carrying a time-dependent coupling weight $w(\sigma, t) \in [0, 1]$.

- For each organ $v \in V(K)$, a *fine complex* $\mathcal{C}_v$ has concepts as vertices and n -ary conceptual bindings as simplices. This is the organ’s internal knowledge.

The stratification answers an ambiguity present in the earliest drafts — *are the vertices concepts or subsystems?* — by answering *both*, at two different levels.

Claims and non-claims. [CONTESTED BY AUDIT] The word “stratified” is used here in a weak, combinatorial sense: two nested scales with a map between them. It does *not* refer to a Whitney stratification, a stratified space in the sense of intersection homology, or a constructible sheaf on a stratified base, and no theorem from those theories is invoked anywhere in this book. Earlier drafts borrowed vocabulary from that literature; the logbook (§0.5) records that most such borrowing was cut as decorative. This is a naming collision, and the reader should not import expectations from it.

2.4 The running example

Every instrument built in this book will be applied to one small complex, so that the same object is measured repeatedly and the measurements can be compared.

Example 2.6 (The complex K_4). Let K_4 have vertex set

$$V = \{\text{S, M, R, V}\}$$

(read: SEARCH, MEMORY, REASONING, VERIFY), with 1-simplices

$$E = \{\{\text{S, M}\}, \{\text{M, R}\}, \{\text{S, R}\}, \{\text{R, V}\}\}$$

and a single 2-simplex $\{\text{S, M, R}\}$, filled. Downward closure holds: all three faces of the triangle are in E .

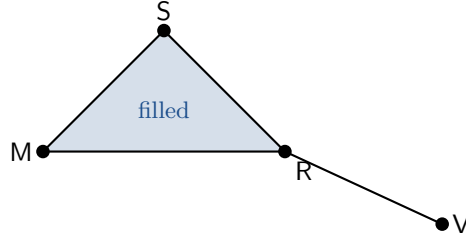


Figure 2.1: The running example K_4 . The triangle $\{S, M, R\}$ is filled: it is one ternary binding, not three binary ones. The edge to V is a pendant.

Its degrees are $\deg S = 2$, $\deg M = 2$, $\deg R = 3$, $\deg V = 1$, and its graph Laplacian, in the vertex order (S, M, R, V) , is

$$L_{K_4} = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}. \quad (2.1)$$

Proposition 2.7 (Spectrum of the running example). **[PROVED]** *The characteristic polynomial of L_{K_4} is $\lambda(\lambda - 1)(\lambda - 3)(\lambda - 4)$, so*

$$\text{spec } L_{K_4} = \{0, 1, 3, 4\}.$$

Proof. Direct expansion of $\det(L_{K_4} - \lambda I)$ gives $\lambda^4 - 8\lambda^3 + 19\lambda^2 - 12\lambda = \lambda(\lambda - 1)(\lambda - 3)(\lambda - 4)$. The eigenvalue 0 is forced: $L_{K_4} \mathbf{1} = 0$ since every row sums to zero. Its multiplicity is 1 because the 1-skeleton is connected (Proposition 4.6). $\square$

The integrality here is a happy accident of this particular graph and should not be expected in general. It is why K_4 was chosen: every quantity in Parts I–IV can be carried through in exact arithmetic.

Chapter 3

The Sheaf

Work plan. **KEEP**

Goal. Keep the sheaf as it stands and add the one thing V2 makes central: that the restriction maps are the *only* place knowledge about wiring can live, so freezing them is not a simplification but a deletion of the architecture.

Raw material. This chapter’s existing text; MATH_TOOLBOX.md I.4 (restriction maps as quantum channels; the potency ladder; Householder representation), I.5 (the measured collapse of a false conflict that identity maps provably cannot produce).

Steps.

1. **Keep the direction argument** (§3.1) in full. Which way the restriction maps point is the chapter’s real content and it is correct.
2. **Add the constant-sheaf refutation early, not late.** Identity restriction maps on a connected graph make the reconciled state the plain average: no knowledge lives in the wiring, and the cohomology goes vacuous. This is **REFUTED** and it is the reason molding exists. It currently appears as a remark; in V2 it belongs near the definition.
3. **State the potency ladder as the chapter’s organising idea.** Level 0 identity (“same thing, verbatim”); level 1 orthogonal (“same content, different basis” — reversible, lossless); level 2 general CPTP (lossy, and the added entropy *is* a measure of connection quality). Give the measured evidence: precision-weighted maps collapse a false conflict (same content, rotated frame) while leaving a real conflict untouched. Identity maps cannot do this, and that asymmetry is the argument.
4. **Record the representation caveat honestly.** Householder products are exactly orthogonal by construction and cheap, but the m -reflection family is *not* closed under the retraction consolidation uses, and reflection-count parity is a genuine topological obstruction — differing parities lie in different components of $O(d)$ and no continuous path joins them. V2 must *choose* a retraction deliberately rather than defaulting. Flag as an open decision.

Done when. A reader can say why identity restriction maps would make the whole architecture decorative, and can state the parity obstruction without consulting the source documents.

Intuition. On top of the shape we lay *data*. Each organ holds a vector describing what it currently believes. Each binding holds a space in which the beliefs of its members are meant to be compared. A sheaf is exactly this: local data on every piece, plus rules for moving data into the places where comparison happens. A “coherent thought” is then a choice of data that agrees everywhere it is compared.

3.1 Which way do the restriction maps point?

Before the definition, one question must be settled, because the source documents answer it in two incompatible ways and the answer determines everything downstream.

Given a face relation $\sigma \leq \tau$ — say a vertex u on an edge e — a sheaf must supply a linear map between $\mathcal{F}(u)$ and $\mathcal{F}(e)$. Which direction?

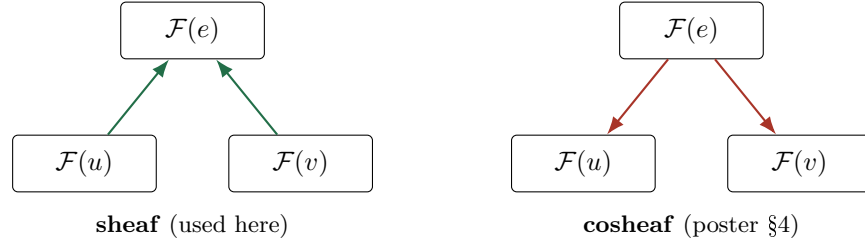


Figure 3.1: The two possible orientations. Only the left one makes $\ker \delta$ the space of agreeing assignments.

Proposition 3.1 (The direction is forced). **[PROVED]** Suppose we want a linear operator δ from vertex data to edge data whose kernel is exactly the set of assignments $x = (x_v)_{v \in V}$ that “agree on every edge”. Then the structure maps must go from faces to cofaces, i.e. $\mathcal{F}_{u \leq e} : \mathcal{F}(u) \rightarrow \mathcal{F}(e)$.

Proof. An assignment gives one vector $x_v \in \mathcal{F}(v)$ per vertex, and nothing on edges. For the two endpoint values x_u, x_v of an edge $e = \{u, v\}$ to be *compared* at all, they must be brought into a common space. The only space available to e is $\mathcal{F}(e)$. Hence we require maps $\mathcal{F}(u) \rightarrow \mathcal{F}(e)$ and $\mathcal{F}(v) \rightarrow \mathcal{F}(e)$, and agreement on e means $\mathcal{F}_{u \leq e}(x_u) = \mathcal{F}_{v \leq e}(x_v)$.

Suppose instead the maps ran downward, $\rho_{e \rightarrow u} : \mathcal{F}(e) \rightarrow \mathcal{F}(u)$. Then from vertex data alone no map into $\mathcal{F}(e)$ exists, so no operator $C^0(\mathcal{C}; \mathcal{F}) \rightarrow C^1(\mathcal{C}; \mathcal{F})$ can be built from the structure maps, and the condition “ x agrees on e ” is not expressible. Downward maps instead assemble into a boundary operator $\partial : C^1(\mathcal{C}; \mathcal{F}) \rightarrow C^0(\mathcal{C}; \mathcal{F})$, whose kernel consists of *edge* data summing to zero at vertices — a flow condition, not an agreement condition. That is the defining structure of a *cellular cosheaf*, and its invariants are homology $H_\bullet$, not the cohomology $H^\bullet$ used throughout this book. $\square$

Claims and non-claims. [REFUTED] The poster section `POSTER/sections/05_sheaf.tex` defines, for $\sigma \subseteq \tau$, a map $\rho_{\tau \rightarrow \sigma} : \mathcal{F}(\tau) \rightarrow \mathcal{F}(\sigma)$, and its diagram draws $\mathcal{F}(e)$ above the vertices with arrows pointing down. It then asserts, three boxes later, that $s \in H^0(\mathcal{C}; \mathcal{F})(\mathcal{C}, \mathcal{F})$ is a global section. By Proposition 3.1 these two statements are incompatible: with downward maps the object defined is a cosheaf and $H^0(\mathcal{C}; \mathcal{F})$ as “the space of compatible assignments” does not arise. The main mathematical description (`DOCS/TATIANA/Mathematical_Description.tex`, Def. 3.1) has the direction correct. The poster is wrong and should be corrected before it is shown to anyone. Logged as Appendix B, item D2.

Exercise 3.2. Take the downward convention seriously for one minute and try to build $\delta : C^0(\mathcal{C}; \mathcal{F}) \rightarrow C^1(\mathcal{C}; \mathcal{F})$ from it. Where exactly does the construction fail — at the definition of δ , or later at the claim $\ker \delta = \{\text{agreeing assignments}\}$? Answering this is the fastest way to internalise why the direction is not a convention.

3.2 The definition

Definition 3.3 (Cellular sheaf). A *cellular sheaf* $\mathcal{F}$ of real inner-product spaces on a simplicial complex $\mathcal{C}$ assigns

- (i) to each simplex $\sigma \in \mathcal{C}$ a finite-dimensional real inner-product space $\mathcal{F}(\sigma)$, the *stalk* over σ ;
- (ii) to each face relation $\sigma \trianglelefteq \tau$ a linear *restriction map*

$$\mathcal{F}_{\sigma \trianglelefteq \tau} : \mathcal{F}(\sigma) \longrightarrow \mathcal{F}(\tau),$$

subject to functoriality: $\mathcal{F}_{\sigma \trianglelefteq \sigma} = \text{Id}$, and for every chain $\sigma \trianglelefteq \tau \trianglelefteq \nu$,

$$\mathcal{F}_{\tau \trianglelefteq \nu} \circ \mathcal{F}_{\sigma \trianglelefteq \tau} = \mathcal{F}_{\sigma \trianglelefteq \nu}. \quad (3.1)$$

Equivalently, $\mathcal{F}$ is a functor from the face poset of $\mathcal{C}$, viewed as a category, to $\mathbf{Vect}_{\mathbb{R}}$.

Remark 3.4 (Functoriality is a real constraint). Condition (3.1) is vacuous on a complex of dimension ≤ 1 , since there are no chains of length two. It becomes a genuine constraint the moment a 2-simplex is filled — which K_4 does. Any procedure that manufactures restriction maps edge-by-edge (for instance by fitting each $\mathcal{F}_{v \trianglelefteq e}$ independently to data) will in general *violate* (3.1) on filled triangles, and the resulting object is then not a sheaf. This failure mode is not hypothetical; it is the standard obstruction encountered when L^2 -projections are used as restriction maps, and the canonical repair is to promote the offending interface to a cell of its own so that the composite is not required to commute. Part VI returns to this.

Definition 3.5 (Cochains). For $k \geq 0$ set

$$C^k(\mathcal{C}; \mathcal{F}) = \bigoplus_{\sigma \in \mathcal{C}_k} \mathcal{F}(\sigma),$$

the space of k -cochains, with the direct-sum inner product $\langle x, y \rangle = \sum_{\sigma} \langle x_{\sigma}, y_{\sigma} \rangle$. An element of $C^0(\mathcal{C}; \mathcal{F})$ is an assignment of one vector to each organ; the current state s of Definition 2.1 lives here.

3.3 What the stalks contain

Construction 3.6 (Deployed stalk data). **[IMPLEMENTED, UNPROVED]** In the deployed system every vertex and edge stalk is $\mathbb{R}^d$ with $d = 384$, and the vector $x_v \in \mathcal{F}(v)$ is a sentence embedding produced by a fixed local encoder. Two invariants are enforced at the level of types:

- (i) dimensions never truncate or pad — a mismatch raises rather than silently reshaping;
- (ii) an idle organ carries *no* stalk value, a formal $\perp$, never the zero vector.

Claims and non-claims. Invariant (ii) is the sheaf-theoretic content of “do not fabricate data”, and it is worth stating why it is not pedantry. The zero vector is a genuine point of $\mathbb{R}^d$: it has a distance to every other point, it contributes to $\|X\|_F^2$, and it participates in every average. Substituting 0 for “unknown” therefore does not represent absence, it asserts a specific belief — that the organ believes the origin — and drags every downstream distance

toward it. The distinction between $\perp$ and 0 is the difference between a measurement not made and a measurement made and found to be zero.

Remark 3.7 (Where the fine complex lives). The logbook occasionally describes concepts as living *inside an organ's stalk*. This is loose. The stalk $\mathcal{F}(v)$ is a 384-dimensional real vector space; the fine complex $\mathcal{C}_v$ is a simplicial complex whose vertices are concepts. They are different objects, and the relation between them is the coarse-graining map π_v of Part III, which sends the concept data of $\mathcal{C}_v$ to a single vector in $\mathcal{F}(v)$. Conflating them makes π_v look like an identity when it is in fact a lossy precision-weighted fusion.

Chapter 4

Coboundary and Laplacian

Work plan. **KEEP**

Goal. Unchanged mathematically. Add the forward references that make it carry V2’s weight, since almost every later chapter reduces to a statement about this operator.

Raw material. This chapter’s existing text; MATH_TOOLBOX.md I.1 ($L = \delta^\top \delta$ real symmetric PSD, which is why the spectral theorem applies for real rather than by analogy; $H^0(\mathcal{C}; \mathcal{F}) = \ker L$).

Steps.

1. Keep the construction and proofs as they are. **KEEP**.
2. Emphasise $H^0(\mathcal{C}; \mathcal{F})(\mathcal{C}; \mathcal{F}) = \ker L$ — “a coherent thought is literally a matrix kernel” — and forward-reference it as the object Chapter 9’s fast flow converges into.
3. Note that L stays symmetric PSD even when restriction maps are general CPTP channels, because such maps are linear on the space of symmetric matrices. This costs nothing and is what lets the potency ladder of Chapter 3 be adopted without redoing the spectral theory.
4. Add a forward reference to the Hodge routing law: the three-way splitting of $C^1(\mathcal{C}; \mathcal{F})$ used throughout V2 is a statement about this coboundary and belongs here in skeleton form.

Done when. The chapter states the two facts every later part relies on ($H^0(\mathcal{C}; \mathcal{F}) = \ker L$; symmetry survives CPTP maps) and points forward to where each is used.

4.1 The coboundary

Definition 4.1 (Coboundary in degree zero). Fix an arbitrary orientation of each edge, writing $e = (u, v)$ for the edge $\{u, v\}$ oriented from u to v . The *coboundary* $\delta : C^0(\mathcal{C}; \mathcal{F}) \rightarrow C^1(\mathcal{C}; \mathcal{F})$ acts edgewise by

$$(\delta x)_e = \mathcal{F}_{v \preceq e}(x_v) - \mathcal{F}_{u \preceq e}(x_u) \in \mathcal{F}(e). \quad (4.1)$$

A 0-cochain x is a *global section* if $\delta x = 0$, that is, if every bound pair agrees after being mapped into its shared comparison space.

Lemma 4.2 (Orientation-independence). **[PROVED]** *Reversing the orientation of an edge e replaces $(\delta x)_e$ by its negative. Consequently $\ker \delta$, the quantity $\|\delta x\|^2$, and the operator $\delta^\top \delta$ are all independent of the chosen orientations.*

Proof. Reversal exchanges the roles of u and v in (4.1), negating the entry. Negating a coordinate leaves its vanishing, its squared norm, and hence the sum of squared norms unchanged. Writing

$\delta' = D\delta$ with D diagonal with entries ± 1 blockwise, we get $\delta'^\top \delta' = \delta^\top D^\top D \delta = \delta^\top \delta$ since $D^\top D = I$. $\square$

Remark 4.3 (Why an arbitrary choice is harmless). Lemma 4.2 is what licenses the phrase “fix an arbitrary orientation”. It is worth checking rather than assuming, because the analogous statement fails for the *signed* per-edge quantity $(\delta x)_e$ itself, which is only defined up to sign. Any diagnostic that reports $(\delta x)_e$ rather than $\|(\delta x)_e\|^2$ is reporting an orientation-dependent quantity and is therefore not well defined.

4.2 The sheaf Laplacian

Definition 4.4 (Sheaf Laplacian and discord). The *degree-zero sheaf Laplacian* is

$$L = \delta^\top \delta : C^0(\mathcal{C}; \mathcal{F}) \rightarrow C^0(\mathcal{C}; \mathcal{F}), \quad (4.2)$$

and the *discord* of a state x is the Dirichlet energy

$$\omega(x) = x^\top Lx = \|\delta x\|^2 = \sum_{e=\{u,v\}} \|\mathcal{F}_{v \leq e}(x_v) - \mathcal{F}_{u \leq e}(x_u)\|^2. \quad (4.3)$$

Proposition 4.5 (Basic properties). **[PROVED]** L is symmetric positive semidefinite, and

$$\omega(x) = 0 \iff \delta x = 0 \iff x \in \ker L.$$

Moreover $\ker \delta = H^0(\mathcal{C}; \mathcal{F})$, the zeroth sheaf cohomology.

Proof. Symmetry: $(\delta^\top \delta)^\top = \delta^\top \delta$. Positive semidefiniteness: $x^\top Lx = \|\delta x\|^2 \geq 0$. If $Lx = 0$ then $\|\delta x\|^2 = x^\top Lx = 0$, so $\delta x = 0$ since $\|\cdot\|$ is a norm; conversely $\delta x = 0$ gives $Lx = 0$ immediately. Finally the cochain complex begins in degree 0 — there are no (-1) -cochains — so $H^0(\mathcal{C}; \mathcal{F}) = \ker \delta^0 / \text{im } \delta^{-1} = \ker \delta$. $\square$

Intuition. ω is a single number: the total squared disagreement across all bindings. It vanishes exactly when the society holds one coherent thought. The point of the construction is that “a coherent thought” has become a precise object — a vector in the kernel of a matrix — rather than a mood.

Proposition 4.6 (Kernel and components). **[PROVED]** If every restriction map is the identity on a common stalk $\mathbb{R}^d$, then

$$L = L_K \otimes I_d,$$

where L_K is the ordinary combinatorial graph Laplacian of the 1-skeleton. Consequently $\text{spec } L$ is $\text{spec } L_K$ with each eigenvalue repeated d times, and $\dim \ker L = d \cdot b_0$, where b_0 is the number of connected components.

Proof. With identity restrictions (4.3) reads $\omega(x) = \sum_{\{u,v\} \in E} \|x_u - x_v\|^2$, which is the Dirichlet form of L_K applied coordinatewise in each of the d components. Hence $L = L_K \otimes I_d$. The spectrum of a Kronecker product with the identity is the spectrum of the factor with multiplicity d . The kernel of L_K is spanned by the indicator vectors of the connected components, of which there are b_0 ; tensoring with $\mathbb{R}^d$ multiplies the dimension by d . $\square$

Claims and non-claims. [IMPLEMENTED, UNPROVED] The deployed system uses identity restriction maps, so Proposition 4.6 describes the regime it actually operates in. This is honest and cheap, but it is the *crudest* possible sheaf: on a connected 1-skeleton, $\ker L$ is the diagonal $\{x : x_u = x_v \ \forall u, v\}$, so the only coherent states are exact consensus and reconciliation is plain averaging. Everything interesting that a sheaf can express — that two organs agree *after* a change of basis, or agree only on a subspace — lives in non-identity restriction maps. Promoting them (learned $\mathcal{F}_{v \leq e}$, for instance by closed-form Procrustes alignment) is the identified route from a diagnostic sheaf to a computational one. It is a direction, not a result.

4.3 The running example, computed

Worked example. Take K_4 from Example 2.6 with identity restrictions and stalk dimension $d = 2$, and give it the state

$$x_S = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_M = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_R = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad x_V = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Read it as: SEARCH and MEMORY agree with each other, REASONING and VERIFY agree with each other, and the two camps are orthogonal.

Discord, edge by edge. Using (4.3) with identity maps, $\omega_e = \|x_u - x_v\|^2$:

$$\begin{aligned} \{S, M\} : \quad & \|(0, 0)\|^2 = 0, \\ \{M, R\} : \quad & \|(1, -1)\|^2 = 2, \\ \{S, R\} : \quad & \|(1, -1)\|^2 = 2, \\ \{R, V\} : \quad & \|(0, 0)\|^2 = 0, \end{aligned} \quad \implies \quad \omega(x) = 4.$$

Localisation. The two guilty edges tie at $\omega_e = 2$. The $\arg \max$ is therefore *not unique*, and a diagnostic that returns a single “worst edge” here is returning an artefact of tie-breaking order, not a measurement. The honest return is the tied set $\{\{M, R\}, \{S, R\}\}$ — which, read back, correctly says that R is the organ at odds with the rest.

Kernel projection. The 1-skeleton is connected, so by Proposition 4.6 the kernel is the diagonal and x_0 is the constant assignment at the mean $\bar{x} = \frac{1}{4}((1, 0) + (1, 0) + (0, 1) + (0, 1)) = (\frac{1}{2}, \frac{1}{2})$. Hence

$$\|x_0\|^2 = 4 \cdot \left(\frac{1}{4} + \frac{1}{4}\right) = 2, \quad x_\perp = x - x_0 = \left(\pm\frac{1}{2}, \mp\frac{1}{2}\right)_v, \quad \|x_\perp\|^2 = 4 \cdot \frac{1}{2} = 2,$$

and Pythagoras checks: $\|X\|_F^2 = 2 + 2 = 4$. □

These numbers are picked up again in Part II, where they are fed to the coherence score ρ and its factorisation, and in Part IV, where the same complex is used to compute b_0 , b_1 and $H^1(\mathcal{C}; \mathcal{F})$.

Exercise 4.7. Recompute $\omega(x)$ for the same complex after filling in a restriction map that is *not* the identity: let $\mathcal{F}_{R \leq e}$ be rotation by 90° for the two edges at R inside the triangle, and the identity elsewhere. Does ω go up or down, and does the resulting assignment of maps

satisfy functoriality (3.1) on the filled triangle $\{S, M, R\}$? The second half of that question is the important one.

Part III

Coherence

Chapter 5

The Coherence Score

Work plan. ¡RECAST!

Goal. Keep the decomposition $\rho = \alpha \cdot \tilde{\rho}$ intact — it is the single highest-value result in the V0 record — while correcting its *status*: in V2 ρ is a diagnostic and a Lyapunov function, never an objective.

Raw material. This chapter’s existing text; MATH.TOOLBOX.md I.2 (the exact decomposition, the Courant–Fischer window, the measured refutation of the “dead dynamic range” worry), I.3 (the weighted Anderson–Morley bound and its stated hypothesis); TATIANA_V1_IDENTITY_MAP.md §8 DON’T 2.

Steps.

1. **Keep the whole decomposition.** $\alpha = \|x^\perp\|^2 / \|X\|_F^2$ (dissent fraction) and $\tilde{\rho}$ (mode index) as a genuine Rayleigh quotient on $(\ker L)^\perp$, hence $\tilde{\rho} \in [\lambda_2(L_K)/B, \lambda_{\max}(L_K)/B]$ by Courant–Fischer — both endpoints computable from the graph *before any data*. ¡KEEP!, [PROVED].
2. **Get the kernel right, in the text, where it can be checked.** $\ker L$ is the *per-component* constants, not the global diagonal. Two earlier write-ups got this wrong, and when $b_0 > 1$ the error scores a between-component separation as dissent when it is in the kernel. State the correct version and note the trap.
3. **Carry the Anderson–Morley hypothesis explicitly.** The bound requires $\|\mathcal{F}_{v \leq e}\| \leq 1$; orthogonal restriction maps suffice, general learned maps do not. An early proof silently assumed identity maps. Any theorem statement must carry the norm bound. [PROVED UNDER STATED CONDITIONS], and say under which conditions.
4. **Retract the novelty framing.** An earlier draft called the weighted bound “the one genuinely new theorem.” It is standard spectral graph theory. Correct this in place rather than quietly dropping it.
5. **Add the V2 status paragraph — this is the recast.** State plainly: *do not minimise ρ* . Perfect coherence is death — the flow drives the state to its harmonic projection and generically to zero. ρ diagnoses and certifies descent; it is never the thing being optimised. The typed diagnosis survives and is the point: high α /low $\tilde{\rho}$ is a structural split, low α /high $\tilde{\rho}$ is one bad edge.
6. **Keep §5.5** (the measured refutation of the dead-dynamic-range worry) as written — it is a clean example of a speculative objection being killed by measurement, and $\alpha = (n-1)(1-\bar{c})/n$ is exact.

Done when. The decomposition is stated with its hypotheses, and no reader could come away thinking ρ is an objective function.

This chapter derives the two numbers the engine computes on every tick, and then shows that one of them is secretly two numbers with different meanings.

5.1 Why a normaliser is needed at all

Intuition. The discord ω has no natural scale. A bigger complex has more edges and therefore a bigger ω for no interesting reason, and a state scaled by a factor of 10 has a discord scaled by 100. What is wanted is a dimensionless thermometer in $[0, 1]$: 0 for perfect agreement, 1 for the maximum disagreement the shape permits. The subtlety is getting it *without* an eigendecomposition, because the score is computed on every tick and an eigensolver is not free.

The natural normaliser is $\lambda_{\max}(L) \|x\|^2$, since $\omega(x) = x^\top Lx \leq \lambda_{\max}(L) \|x\|^2$ by the Rayleigh bound. But computing $\lambda_{\max}$ is exactly what we are trying to avoid. The way out is a classical degree bound.

Theorem 5.1 (Anderson–Morley). **[PROVED]** *Let G be a finite simple graph with combinatorial Laplacian L_G . Then*

$$\lambda_{\max}(L_G) \leq \max_{\{u,v\} \in E} (\deg u + \deg v) =: B. \quad (5.1)$$

Proof. Write $L_G = D - A$ and let $Q = D + A$ be the signless Laplacian. Because $Q = NN^\top$ for N the unsigned incidence matrix, Q is positive semidefinite, and $\lambda_{\max}(L_G) \leq \lambda_{\max}(Q)$: passing to the bipartite double cover exchanges L_G and Q while preserving the spectral radius, and L_G is a principal restriction of that cover’s signless Laplacian.

It therefore suffices to bound $\lambda_{\max}(Q)$. Let y be a unit eigenvector of Q for $\lambda_{\max}$, let u maximise $|y_u|$, and let v be a neighbour of u maximising $|y_v|$. From $\lambda_{\max} y_u = \deg(u) y_u + \sum_{w \sim u} y_w$ we get

$$(\lambda_{\max} - \deg u) |y_u| \leq \deg(u) |y_v|,$$

and symmetrically $(\lambda_{\max} - \deg v) |y_v| \leq \deg(v) |y_u|$. Multiplying the two inequalities and cancelling $|y_u| |y_v| > 0$ gives $(\lambda_{\max} - \deg u)(\lambda_{\max} - \deg v) \leq \deg u \deg v$, i.e. $\lambda_{\max}^2 - \lambda_{\max}(\deg u + \deg v) \leq 0$, whence $\lambda_{\max} \leq \deg u + \deg v \leq B$. $\square$

Remark 5.2 (The bound is loose, and the looseness costs something). (5.1) is tight only on special graphs — it is exact on complete bipartite graphs. On the running example K_4 it gives $B = 5$ while $\lambda_{\max} = 4$, a 25% over-estimate. Over-bounding $\lambda_{\max}$ compresses ρ toward zero, which is one of three contributors to the dead dynamic range documented in Section 5.5. Finding a tighter eigensolver-free bound is Open Problem 22.4.

5.2 The score

Definition 5.3 (Coherence score). Let $X \in \mathbb{R}^{|V| \times d}$ stack the active stalk vectors as rows, with Frobenius norm $\|X\|_F$. In the identity-restriction regime the *coherence score* is

$$\rho = \frac{\omega}{B \|X\|_F^2} \quad (5.2)$$

When $E = \emptyset$ or $\|X\|_F = 0$, the score is *undefined* and returns $\perp$ — never 0 and never 1.

Proposition 5.4 ($\rho \in [0, 1]$). **[PROVED UNDER STATED CONDITIONS]** *With identity restriction maps, $0 \leq \rho \leq 1$.*

Proof. Lower bound: $\omega \geq 0$ and $B \|X\|_F^2 > 0$ whenever the score is defined. Upper bound: by Proposition 4.6, $L = L_K \otimes I_d$, so $\lambda_{\max}(L) = \lambda_{\max}(L_K) \leq B$ by Theorem 5.1. Hence $\omega = x^\top Lx \leq \lambda_{\max}(L) \|x\|^2 \leq B \|X\|_F^2$, using $\|x\| = \|X\|_F$ under the row-stacking identification. $\square$

Claims and non-claims. The hypothesis in Proposition 5.4 is not cosmetic. With *non-identity* restriction maps the Kronecker factorisation fails, $\lambda_{\max}(L)$ is no longer the spectrum of any $|V| \times |V|$ matrix, and B need not bound it. The correct general hypothesis is an operator-norm condition: if every $\|\mathcal{F}_{v \leq e}\| \leq 1$ then a weighted version of Theorem 5.1 still applies. Without it the score can exceed 1. The recorded design decision is that this case is *reported*, not silently clamped — a value $\rho > 1$ is evidence that the normalisation hypothesis has been violated, and truncating it destroys the only evidence there is.

Remark 5.5 (Why $\perp$ is not 0). An empty edge set means no comparison was made. Returning 0 would report perfect agreement, which is a measurement; returning $\perp$ reports that no measurement exists. The distinction propagates: the controller of Chapter 18 carries a third mode precisely so that “no measurement” does not default to “all is well”.

5.3 The score answers two questions at once

Intuition. The numerator ω is *blind* to consensus, because consensus is exactly $\ker L$ and the kernel contributes nothing to the quadratic form. The denominator $\|X\|_F^2$ counts consensus in full. So ρ is disagreement divided by (disagreement plus consensus), tangled together with a second and quite different quantity saying which *shape* the disagreement takes. Untangling them costs one orthogonal projection.

Proposition 5.6 (Exact factorisation). **[PROVED UNDER STATED CONDITIONS]** *Assume identity restrictions. Let $x = x_0 \oplus x_\perp$ be the orthogonal decomposition with x_0 the projection onto $\ker L$, and suppose $x_\perp \neq 0$. Then $\omega(x) = x_\perp^\top Lx_\perp$, $\|X\|_F^2 = \|x_0\|^2 + \|x_\perp\|^2$, and*

$$\rho = \underbrace{\frac{\|x_\perp\|^2}{\|X\|_F^2}}_{\alpha} \cdot \underbrace{\frac{\omega}{B\|x_\perp\|^2}}_{\tilde{\rho}} \quad (5.3)$$

where α , the dissent fraction, says how much of the state is not consensus, and $\tilde{\rho}$, the mode index, says which mode the disagreement occupies.

Proof. Since $Lx_0 = 0$ and L is symmetric, the cross terms in $x^\top Lx = (x_0 + x_\perp)^\top L(x_0 + x_\perp)$ vanish, because $x_0^\top Lx_\perp = (Lx_0)^\top x_\perp = 0$. The norm identity is Pythagoras. The factorisation is multiplication and division of (5.2) by $\|x_\perp\|^2$. $\square$

Remark 5.7 (The kernel is per-component, not the global mean). For a connected 1-skeleton, $\ker L$ is the diagonal and $x_0 = \mathbf{1} \otimes \bar{x}$ with $\bar{x}$ the global mean. For $b_0 > 1$ this is *false*: by Proposition 4.6 the kernel consists of assignments constant on *each component*, so x_0 subtracts the **per-component** mean. Using the global mean when $b_0 > 1$ scores between-component separation as dissent, when in fact it lies *inside* the kernel and is not disagreement at all. This is a correctness condition, not an optimisation.

Proposition 5.8 (The achievable window). **[PROVED UNDER STATED CONDITIONS]** $\tilde{\rho} = R_L(x_\perp)/B$ is a Rayleigh quotient of L restricted to $(\ker L)^\perp$, so by Courant–Fischer

$$\tilde{\rho} \in \left[\lambda_{\min}^+(L_K)/B, \lambda_{\max}(L_K)/B \right], \quad (5.4)$$

where $\lambda_{\min}^+$ is the smallest nonzero eigenvalue — the Fiedler value when K is connected. Both endpoints are graph invariants, computable from the 1-skeleton before any data is seen, and needing recomputation only when the topology changes.

Proof. $x_\perp$ ranges over $(\ker L)^\perp$, an invariant subspace on which L is positive definite with spectrum the nonzero eigenvalues. The Rayleigh quotient of a symmetric operator on an invariant subspace attains exactly the closed interval between the extreme eigenvalues there. Dividing by the constant B gives (5.4). $\square$

Claims and non-claims. Four points, in order of importance.

(i) (5.3) resolves the non-monotonicity of Remark 18.2: adding a well-aligned organ inflates $\|X\|_F^2$ and barely moves $\|x_\perp\|^2$, so α dilutes while $\tilde{\rho}$ is untouched. The controller should gate on $\tilde{\rho}$ and report α alongside.

(ii) (5.4) is **normalisation, not derivation**. Setting a threshold ε as a quantile q of the window still requires choosing q . What changes is that ε is expressed in units the graph supplies and adapts automatically to rewiring. It is a magic number made topology-relative, not a magic number eliminated, and it must not be sold as more than that.

(iii) The split is exact *only* for identity restrictions. With non-identity maps, $\ker L$ is not the per-component constants and no $|V| \times |V|$ matrix carries the spectrum of L . The correct behaviour is then to report α , $\tilde{\rho}$ and the window as $\perp$ rather than approximate them. ρ itself is unaffected.

(iv) When $x_\perp = 0$ exactly, $\alpha = 0$ and $\tilde{\rho} = \perp$: perfect consensus has no mode. Returning 0 for $\tilde{\rho}$ would assert “the most global mode possible”, a measurement that was not made.

5.4 The running example, continued

Worked example. Continue Example 2.6 with the state of Chapter 4, where we found $\omega = 4$, $\|X\|_F^2 = 4$ and $\|x_\perp\|^2 = 2$.

The degrees are 2, 2, 3, 1, so

$$B = \max\{2+2, 2+3, 2+3, 3+1\} = 5.$$

Hence

$$\rho = \frac{4}{5 \cdot 4} = \frac{1}{5}, \quad \alpha = \frac{2}{4} = \frac{1}{2}, \quad \tilde{\rho} = \frac{4}{5 \cdot 2} = \frac{2}{5},$$

and the factorisation checks: $\alpha \cdot \tilde{\rho} = \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{5} = \rho$.

By Proposition 2.7 the nonzero spectrum is $\{1, 3, 4\}$, so

$$\text{window} = \left[\frac{1}{5}, \frac{4}{5} \right] = [0.2, 0.8],$$

and $\tilde{\rho} = 0.4$ sits at window position $(0.4 - 0.2)/(0.8 - 0.2) = \frac{1}{3}$.

Reading the pair. $\alpha = \frac{1}{2}$ is large: half the state’s energy is not consensus. $\tilde{\rho}$ at $\frac{1}{3}$ of its window is low. By the typed reading of Remark 18.3, large α with small $\tilde{\rho}$ is a *two-camp split* — which is exactly how the state was built. The single number $\rho = 0.2$ reads as “mild disagreement” and misses this completely. $\square$

Exercise 5.9. Add a fifth organ P attached only to S, with $x_P = x_S = (1, 0)$: a perfectly aligned new organ contributing no disagreement. Recompute B , ρ , α and $\tilde{\rho}$. Which changed, and does the change match the dilution claim in item (i) above?

5.5 The dead dynamic range

Claims and non-claims. [REFUTED] (measured 2026-07-28). It was claimed that the compressed dynamic range of ρ is caused by encoder anisotropy acting through α . The claim does not survive. The deployed encoder returns unit-normalised vectors, for which

$$\alpha = \frac{(n-1)(1-\bar{c})}{n}, \quad (5.5)$$

with $\bar{c}$ the mean pairwise cosine. At the observed $\bar{c} \approx 0.449$ and $n = 7$ this gives $\alpha \approx 0.46$, which is not small. The compression is produced *jointly* by α , by $\tilde{\rho}$ sitting mid-window, and by B over-bounding $\lambda_{\max}$. On a seven-organ complex the measurement was $\tilde{\rho} \approx 0.47$ inside a window $[0.146, 0.854]$: the mode index has healthy dynamic range that ρ was multiplying away.

Exercise 5.10. Derive (5.5). Assume $\|x_v\| = 1$ for all v , take x_0 to be the constant assignment at the mean, and expand $\|x_\perp\|^2 = \sum_v \|x_v - \bar{x}\|^2$ in terms of pairwise inner products. The identity is three lines, and it is why the refutation is airtight rather than merely suggestive.

5.6 The precision-weighted generalisation

Construction 5.11 (Weighted discord and normaliser). **[IMPLEMENTED, UNPROVED]** Give each edge a *precision* $\pi_e > 0$ and set

$$\omega_\pi(x) = \sum_{e=\{u,v\}} \pi_e \|\mathcal{F}_{v \leq e}(x_v) - \mathcal{F}_{u \leq e}(x_u)\|^2, \quad (5.6)$$

the Dirichlet form of $L_\pi = \delta^\top P \delta$ with $P = \text{diag}(\pi_e)$. Define the precision-weighted degree and the weighted normaliser

$$d_\pi(v) = \sum_{e \ni v} \pi_e, \quad B_\pi = \max_{e=\{u,v\}} (d_\pi(u) + d_\pi(v)),$$

and replace B by B_π throughout.

Proposition 5.12 (The weighted bound). **[PROVED]** *In the identity-restriction regime, $\lambda_{\max}(L_\pi) \leq B_\pi$.*

Proof. For rational weights, the weighted graph Laplacian is the unweighted Laplacian of the multigraph in which edge e is replaced by π_e parallel copies; Theorem 5.1 applied there gives the bound with degrees replaced by weighted degrees. The general case follows because both sides are continuous in the weights and the rationals are dense. $\square$

Claims and non-claims. This is where organ-level confidence legitimately enters the architecture. It does *not* enter as a stalk variance: the dimensional analysis of Chapter 6 shows a confidence-derived stalk floor is d -extensive and swamps everything else at $d = 384$. As an edge precision, entering as $s_e = -\ln c_e/d$, its trace contribution is $O(1)$ and it is dimensionally harmless. The same number is load-bearing in one place and destructive in the other, and the entire difference is where in the sheaf it is attached.

Part IV

Semantic Geometry

Chapter 6

Semantic Geometry

Work plan. **KEEP**

Goal. Keep the geometry and make its dispatch rule — which metric for which job — the chapter’s organising principle rather than a remark.

Raw material. This chapter’s existing text; `MATH_TOOLBOX.md` IV.1 (the estimation/comparison dispatch), IV.2 (Cone–Bures: kept as a metric, its closeness claim refuted), IV.3 (what a concept is), I.6 (the variance floor — the largest measured downstream effect in the V0 record).

Steps.

1. **Lead with the dispatch rule.** ESTIMATION uses information/Fisher–Rao geometry, where a theorem exists (precision- weighted fusion is a genuine MLE); COMPARISON uses Bures–Wasserstein transport geometry, where a modelling choice exists. They genuinely disagree — by Agueh–Carlier the Wasserstein barycenter of Gaussians has the *plain* weighted mean, not the precision-weighted one. Never blend comparison into estimation.
2. **Own the seam.** π_v fuses in information geometry but its output is immediately compared in transport geometry by ω ; π_v is *not* the transport barycenter of its own fine complex. This is a paragraph the book owes a reader, not a bug. Write it.
3. **Record the rejected repair.** Training a blend weight between the two geometries is **[RE-FUTED]**, on four grounds worth keeping because they generalise: it replaces a choice-with-a-reason by a fitted-constant-with- none; no training signal exists; one scalar wants a sweep, not an optimizer; and the two uses are asymmetric, so blending damages a proven MLE for a fitted number.
4. **Upgrade what a concept is.** A Gaussian stalk’s level sets are ellipsoids, hence convex, so the SPD stalk already *is* a convex conceptual region in Gärdenfors’ sense — concepts have extent, and treating one as a point discards exactly the structure this geometry builds. State this as the chapter’s conclusion.
5. **Keep the variance-floor fix and its numbers.** The correct invariant is “floor carries trace $O(1)$ ” and not “shared epsilon”; getting this wrong once cost a $47\times$ slowdown, and fixing it moved the epistemic term from 229.76 to 0 and made the transport term non-inert. **[MEASURED]**. It is the best worked example in the book of a hardcoded constant silently flattening a whole geometry.
6. **Keep Cone–Bures with its refutation attached.** It survives as a genuine metric (proved, measured, bounded, and it deletes a hand-set threshold). Its “tracks true Hellinger–Kantorovich to under 1%” claim does *not* survive real text. Keep both facts adjacent; report the diagnostic ratio as telemetry and never as a claim.

Done when. A reader can state which metric to use for a given job and why, and can explain the

seam between them without treating it as an oversight.

Part I built the shape and Part II measured disagreement across it. This part supplies the geometry *inside* an organ: what a concept is, how many concepts collapse into the single vector a stalk holds, and how far apart two uncertain concepts are.

6.1 A concept is a Gaussian

Intuition. A concept is not a point. It is a point plus an admission of how uncertain that point is, and — more subtly — of the *directions* in which it is uncertain. A concept may be sharply located along one axis of meaning and vague along another. A Gaussian with a structured covariance is the smallest object that records all three.

Definition 6.1 (Concept). A *concept* is a Gaussian $\mathcal{N}(\mu_i, \Sigma_i)$ on $\mathbb{R}^d$ with low-rank-plus-isotropic covariance

$$\Sigma_i = U_i U_i^\top + D_i I_d, \quad U_i \in \mathbb{R}^{d \times k_i}, \quad D_i > 0. \quad (6.1)$$

Here μ_i is the embedding, $U_i U_i^\top$ is anisotropic learned spread, and D_i is an isotropic noise floor — the epistemic humility of the concept.

The rank k_i is kept small (in practice $k_i \leq 8$). This is not a performance compromise: it is what makes every computation below reduce to a $k \times k$ problem rather than a $d \times d$ one, and it is the reason the geometry is affordable at all.

6.2 The coarse-graining π_v

Intuition. An organ holds many concepts but exposes one vector to the coarse complex. Which vector? Not the plain average: a near-certain concept should dominate a vague one. The right notion of weight is *precision* — inverse variance — and the resulting formula is exactly how independent noisy measurements of a single quantity are optimally combined.

Construction 6.2 (The coarse-graining map). For an organ v with active concepts $\{\mathcal{N}(\mu_i, \Sigma_i)\}_{i=1}^n$, the coarse stalk value is

$$\pi_v = \left(\sum_{i=1}^n \Sigma_i^{-1} \right)^{-1} \left(\sum_{i=1}^n \Sigma_i^{-1} \mu_i \right) \quad (6.2)$$

If v has no active concept then $\pi_v = \perp$, and v is excluded from the computation of ρ rather than contributing a zero vector.

Proposition 6.3 (π_v is the maximum-likelihood fusion). **[PROVED]** Suppose $\theta \in \mathbb{R}^d$ is a latent vector and each concept is an independent observation $\mu_i = \theta + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \Sigma_i)$. Then the maximum-likelihood estimate of θ is (6.2).

Proof. The log-likelihood is $\ell(\theta) = -\frac{1}{2} \sum_i (\mu_i - \theta)^\top \Sigma_i^{-1} (\mu_i - \theta) + \text{const}$. Its gradient is $\nabla_\theta \ell = \sum_i \Sigma_i^{-1} (\mu_i - \theta)$, and setting it to zero gives $(\sum_i \Sigma_i^{-1}) \theta = \sum_i \Sigma_i^{-1} \mu_i$. The matrix $\sum_i \Sigma_i^{-1}$ is positive definite (each Σ_i^{-1} is, since $D_i > 0$), hence invertible, and ℓ is strictly concave, so the stationary point is the unique maximum. $\square$

Remark 6.4 (Three properties worth naming). The fusion (6.2) is *order-independent* (both sums are commutative), *non-destructive* (every concept contributes to both the precision sum and the mean; none is discarded), and *n-ary* rather than iterated-pairwise. These are not incidental: an order-dependent or lossy summary would make the coarse measurement depend on the sequence in which an organ happened to acquire its concepts.

Two computational paths, both exact

- **Isotropic fast path.** When all U_i are empty, $\Sigma_i^{-1} = D_i^{-1}I$ and (6.2) collapses to the scalar precision-weighted average

$$\pi_v = \frac{\sum_i D_i^{-1} \mu_i}{\sum_i D_i^{-1}}, \quad (6.3)$$

computable in $O(nd)$ with no matrix inversion.

- **General path.** With $U_i \neq \emptyset$, invert each Σ_i by the Sherman–Morrison–Woodbury identity

$$\Sigma_i^{-1} = D_i^{-1}I - D_i^{-1}U_i(I_{k_i} + D_i^{-1}U_i^\top U_i)^{-1}U_i^\top D_i^{-1}, \quad (6.4)$$

accumulate into a dense information matrix, and solve once, at $O(ndk + d^3)$.

Lemma 6.5 (Woodbury for this covariance). **[PROVED]** (6.4) holds for $\Sigma = UU^\top + DI$ with $D > 0$.

Proof. The Woodbury identity states $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$. Take $A = DI$, $C = I_k$, $V = U^\top$. Then $A^{-1} = D^{-1}I$ and $C^{-1} + VA^{-1}U = I_k + D^{-1}U^\top U$, which is positive definite and hence invertible. Substituting gives (6.4). The cost is dominated by the $k \times k$ inverse, not by anything of size d . $\square$

Remark 6.6 (The floor is not optional). A lower bound $D_i \geq 10^{-6}$ is imposed. Without it, a concept reported as near-certain produces $1/D_i \approx 10^{16}$ and the information matrix loses all contribution from every other concept to rounding. The floor pins π_v strongly but finitely. Numerically the general path was checked against a dense $(\sum_i \Sigma_i^{-1})^{-1}$ solve and agreed to 2.2×10^{-16} ; equal precisions reduce to the plain mean, and a 9:1 precision ratio pulls π_v to the confident concept, as required.

Remark 6.7 (This map is the two-level glue). After execution, each organ's coarse stalk is *overwritten* by its own π_v . That is what makes the coarse ρ and the fine geometry one measurement at two resolutions rather than two unrelated numbers. It also means π_v is lossy in a load-bearing way: whatever the fine complex knew that does not survive (6.2) is invisible to every coarse diagnostic.

6.3 Distance between uncertain concepts

Intuition. How far apart are two concepts? Comparing centres alone throws away the spreads. The 2-Wasserstein — optimal-transport — distance between Gaussians accounts for both, and is a genuine metric rather than a heuristic score.

Definition 6.8 (2-Wasserstein between Gaussians).

$$W_2^2(\mathcal{N}(\mu_1, \Sigma_1), \mathcal{N}(\mu_2, \Sigma_2)) = \|\mu_1 - \mu_2\|^2 + \text{Tr } \Sigma_1 + \text{Tr } \Sigma_2 - 2 \text{Tr} \left[(\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2})^{1/2} \right]. \quad (6.5)$$

The covariance part alone is the squared Bures–Wasserstein distance d_{BW}^2 on $\text{SPD}(d)$.

Proposition 6.9 (Isotropic reduction to $O(k^3)$). **[PROVED]** *If $\Sigma_1 = D_1 I$ is isotropic and $\Sigma_2 = U_2 U_2^\top + D_2 I$ with $U_2 \in \mathbb{R}^{d \times k}$, then, writing $\{\sigma_j^2\}_{j=1}^k$ for the eigenvalues of the $k \times k$ matrix $U_2^\top U_2$,*

$$W_2^2 = \|\mu_1 - \mu_2\|^2 + (\text{Tr } \Sigma_1 + \text{Tr } \Sigma_2) - 2\sqrt{D_1} \left[\sum_{j=1}^k \sqrt{\sigma_j^2 + D_2} + (d - k)\sqrt{D_2} \right]. \quad (6.6)$$

Proof. Since $\Sigma_1^{1/2} = \sqrt{D_1} I$, we get $\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2} = D_1 \Sigma_2$ and hence $(D_1 \Sigma_2)^{1/2} = \sqrt{D_1} \Sigma_2^{1/2}$, so the cross term is $2\sqrt{D_1} \text{Tr } \Sigma_2^{1/2}$. The eigenvalues of $\Sigma_2 = U_2 U_2^\top + D_2 I$ are $\sigma_j^2 + D_2$ on $\text{im } U_2$ and D_2 with multiplicity $d - k$ on $\ker U_2^\top$, because the nonzero eigenvalues of $U_2 U_2^\top$ coincide with those of $U_2^\top U_2$. Summing $\sqrt{\lambda}$ over the spectrum gives (6.6). Only a $k \times k$ eigendecomposition is needed. $\square$

This is exact for isotropic Σ_1 , not a diagonal approximation.

6.4 The two terms are not commensurable

Construction 6.10 (Reporting the split). W_2^2 is a sum of two quantities measuring different things, and they are returned separately rather than added:

$$W_2^2 = \underbrace{\|\mu_1 - \mu_2\|^2}_{\text{semantic: what they say}} + \underbrace{\text{Tr } \Sigma_1 + \text{Tr } \Sigma_2 - 2 \text{Tr} [(\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2})^{1/2}]}_{\text{epistemic: how sure they are}}. \quad (6.7)$$

Proposition 6.11 (Degeneration in the isotropic regime). **[PROVED]** *If both concepts are isotropic, $\Sigma_i = D_i I$ with U_i empty, then*

$$W_2^2 = \|\mu_1 - \mu_2\|^2 + d(\sqrt{D_1} - \sqrt{D_2})^2. \quad (6.8)$$

In particular $D_1 = D_2$ implies $W_2^2 = \|\mu_1 - \mu_2\|^2$ exactly.

Proof. Set $k = 0$ in Proposition 6.9. Then $\text{Tr } \Sigma_1 + \text{Tr } \Sigma_2 = d(D_1 + D_2)$ and the cross term is $2\sqrt{D_1} \cdot d\sqrt{D_2}$, so the covariance part equals $d(D_1 + D_2 - 2\sqrt{D_1 D_2}) = d(\sqrt{D_1} - \sqrt{D_2})^2$. $\square$

Claims and non-claims. Proposition 6.11 recorded two uncomfortable facts. Both were true when written; both were subsequently repaired. The diagnosis is retained because it is what forced the repair and because the structural lesson survives it.

(i) *The machinery was inert.* Every grown concept carried the same prior $D = 1.0$ with empty U , so $D_1 = D_2$ and W_2^2 was *exactly* squared Euclidean distance — an optimal-transport apparatus with a Wasserstein-shaped proof attached and nothing to do. After the repair, D is the shrinkage floor of Remark 6.15 and varies with effective sample size and with rank, so $D_1 \neq D_2$ in general. On two stalks with equal means and ranks 0 and 1, the epistemic term moves from 0 to 0.206: the geometry is live.

(ii) *It would have become actively harmful the moment calibration was fixed.* The se-

mantic term is *d-intensive* — bounded by 4 for unit-normalised embeddings, whatever d is — while the epistemic term is *d-extensive*. At $d = 384$, a prior $D_1 = 1.0$ against a confident concept with $D_2 = -\ln 0.95$ gave an epistemic term of roughly 230 against a semantic term of 2: about 115 to 1, driven entirely by a self-reported confidence that the architecture elsewhere declines to trust. Merge, split and consolidation would then have been governed by hallucinated confidences. This is a latent defect that switches on precisely when the system is *improved*, which is the worst kind.

The lesson survives both fixes and is why the terms are reported apart: summing a *d-intensive* and a *d-extensive* quantity and thresholding the sum is a dimensional error, not a modelling preference.

Exercise 6.12. Verify the 115:1 figure. Take $d = 384$, $D_1 = 1.0$, $D_2 = -\ln 0.95$, both isotropic, and evaluate (6.8). Then repeat with the shrinkage floor $D = O(1/d)$ of Remark 6.15 and confirm the epistemic term drops to order 1. The point of doing this by hand is that the failure is visible in the exponents, before any numbers are substituted.

6.5 Fusion and its entropy

Definition 6.13 (Precision-weighted fusion). The fusion of two concepts is the information-form product

$$\Lambda_i = \Sigma_i^{-1}, \quad \Sigma_\star = (\Lambda_1 + \Lambda_2)^{-1}, \quad \mu_\star = \Sigma_\star(\Lambda_1\mu_1 + \Lambda_2\mu_2), \quad (6.9)$$

which is (6.2) for $n = 2$. Full covariances are fused via Woodbury (6.4), never element-wise.

Proposition 6.14 (Fused entropy by the matrix determinant lemma). **[PROVED]** *The differential entropy $h = \frac{1}{2} \ln((2\pi e)^d \det \Sigma_\star)$ is computable from a $k \times k$ determinant: for $\Sigma = UU^\top + DI$,*

$$\det \Sigma = D^d \det(I_k + D^{-1}U^\top U). \quad (6.10)$$

Proof. The matrix determinant lemma gives $\det(A + UV^\top) = \det A \cdot \det(I + V^\top A^{-1}U)$. With $A = DI$ and $V = U$ we have $\det A = D^d$ and $V^\top A^{-1}U = D^{-1}U^\top U$. $\square$

Remark 6.15 (Where confidence actually goes). **[IMPLEMENTED, UNPROVED]** An earlier design set the noise floor from the generating model's self-reported confidence via $D = -\ln c$, with a named prior $D = 1.0$ when c was unavailable. Both were removed. The floor is now a shared shrinkage floor

$$D = \varepsilon + \frac{\kappa}{d(n_{\text{eff}} + \kappa)} = O(1/d),$$

independent of any reported confidence. The scaling is the whole point: D feeds a *d-extensive* Bures term, so a floor of order 1 contributes a trace of order d and swamps a semantic term bounded by 4. At $d = 384$ the two retired forms give isotropic traces of about 19.7 and 384 respectively; the shrinkage floor gives 0.88.

Confidence is *relocated, not discarded*: its home is the edge precision of Construction 5.11, entering as $s_e = -\ln c_e/d$ so that its trace contribution is $O(1)$. There it is dimensionally harmless and load-bearing; as an isotropic stalk variance it was neither. Against the deployed model, token log-probabilities are unavailable, so c is genuinely unknown and the engine warns once rather than fabricating a value. This is why ρ , and not self-reported confidence, is the primary signal.

Claims and non-claims. The pattern in this chapter is worth extracting, because it recurs. Each of the three defects — inert geometry, dimensional mismatch, fabricated confidence — was invisible to every test that checked whether the code *ran*, and visible immediately to a check of whether the quantities being combined had the same units. Dimensional analysis is the cheapest correctness tool available here and it was applied late.

Part V

Cohomology and the Growth Address

Chapter 7

Cohomology and the Diagnostic Hierarchy

Work plan. **¡RECAST!**

Goal. This is the chapter V2 changes most in *role*. In V0 the cohomology was a diagnostic hierarchy. In V2 the Hodge decomposition is the routing law of the wake phase and the harmonic component is the growth address — so this chapter carries the architecture, not commentary on it.

Raw material. This chapter’s existing text (§“the hierarchy is vacuous as currently fed” is the key surviving content); `MATH_TOOLBOX.md` III.1 (Proposition 8.2 and the reframe it forced), III.2 (measuring a genuine η : four runs, one retraction), III.6 (the b_1 -swamping bug and the triangle rule); `TATIANA_V1_IDENTITY_MAP.md` §4.1, §9.

Steps.

1. **Open with Proposition 8.2, not with the hierarchy.** $[\delta x] = 0$ in $H^1(\mathcal{C}; \mathcal{F})$ *unconditionally* — a coboundary is by definition not a gluing failure. This holds for every state, every sheaf, identity restrictions or not. It is trivial to prove and it is the deepest correct criticism the project received: the engine’s only 1-cochain is always exact, so growth has no address, structurally, no matter how the state is computed. **[PROVED]**. Everything else in the chapter is downstream of it.
2. **State what it kills.** The original growth intuition (read the address off $H^1(\mathcal{C}; \mathcal{F})$ of a state the system already computes) is dead permanently. So is the “residuals of learned restriction maps” repair, which is **[REFUTED]** for the identical reason — those residuals are still coboundaries of the learned maps, still exact.
3. **Give the fix as the reframe it is.** Growth needs a genuinely *measured* 1-cochain η that is not the coboundary of any single global state, plus the Hodge split; the harmonic component is the address. State this as the pivot of the whole architecture.
4. **Flag the V2-specific hole loudly.** In V0, η came from organs emitting pairwise judgements that disagreed. *V2 has no organs*. Where a genuinely non-exact η comes from for a single kernel is **open** and is the most important unanswered question in the book. Do not paper over it; give it its own section and forward-reference Chapter 22.
5. **Keep the elicitation lesson.** A symmetric $\{u, v, \text{agreement}\}$ schema *cannot* be a 1-cochain — antisymmetrising a symmetric table gives $\eta \equiv 0$ identically. Any future source of η , organ-based or not, must be antisymmetric by construction. Two earlier designs failed informatively before one worked.
6. **Keep the dimensionality trap.** A 1-dimensional harmonic subspace can report *that* an obstruction exists but never *where* — at $b_1 = 1$ every harmonic component is a multiple of one

fixed vector, so the “consistent growth address” it reports is an artifact. Any configuration used for a real address needs $b_1 \geq 2$. This nearly became a reported finding and must stay visible.

7. **Keep the triangle rule and the swamping bug.** Record a 2-simplex exactly when its three concepts co-fire; the earlier claim that “no triangles $\Rightarrow b_1 = 0$ ” was backwards — no triangles *maximises* b_1 , and inserting each assembly as a clique manufactures $(n-1)(n-2)/2$ artifact cycles that swamp the real address. Also record that $b_1 = E - V + b_0 - F$ is *wrong* once triangles share edges; compute b_1 from ranks, always.
8. **Correct the headline statistic.** On a sparse graph a high “holes” fraction is the *null expectation* ($\mathbb{E} = b_1/|E|$), not evidence — measured twelve sigma below a null in one run. The usable output is per-cycle *localisation* of harmonic mass, which is what Construction 6 actually consumes. Replace the fraction with localisation wherever the old text reports it.

Done when. A reader can prove Proposition 8.2 unaided, explain why it makes the growth address the central problem, and state what would have to be true of a source of η for it to work — including that V2 does not yet have one.

7.1 Two pathologies that must not be confused

Definition 7.1 (Sheaf cohomology in low degrees). With coboundaries $\delta^k : C^k(\mathcal{C}; \mathcal{F}) \rightarrow C^{k+1}(\mathcal{C}; \mathcal{F})$,

$$H^0(\mathcal{C}; \mathcal{F}) = \ker \delta^0 \quad (\text{global sections}), \quad H^1(\mathcal{C}; \mathcal{F}) = \ker \delta^1 / \text{im } \delta^0 \quad (\text{gluing failure}).$$

Intuition. A *structural hole* — $b_1 \neq 0$, a Betti number of the bare shape — means a *binding is missing*: no theorem yet ties some loop together. A *gluing failure* — $H^1(\mathcal{C}; \mathcal{F}) \neq 0$ of the sheaf — means the data are *contradictory*: the organs cannot be reconciled. The first calls for growth; the second calls for repair. Treating a contradiction as a hole makes you add structure when you should resolve a conflict, and treating a hole as a contradiction makes you argue when you should build.

Remark 7.2 (b_1 is earned here). b_1 is a defensible diagnostic *only* because simplices were defined to mean genuine n -ary binding (Remark 2.3). A filled 2-simplex really asserts a joint relation, so an unfilled 1-cycle really is a missing joint relation. Most applications of topological data analysis to knowledge graphs skip this justification and compute b_1 of a complex whose 2-cells mean nothing in particular, at which point the number is an artefact of the construction rather than a fact about the knowledge.

7.2 The hierarchy is vacuous as currently fed

This is the most important negative result in the book, and it is worth stating as a proposition rather than as a caveat.

Proposition 7.3 (Exactness of the emitted cochain). **[PROVED]** *Suppose the 1-cochain submitted for cohomological analysis is obtained as $y = \delta^0 x$ for some global state $x \in C^0(\mathcal{C}; \mathcal{F})$. Then $[y] = 0$ in $H^1(\mathcal{C}; \mathcal{F})$, for every x , every sheaf $\mathcal{F}$, and every choice of restriction maps.*

Proof. $H^1(\mathcal{C}; \mathcal{F}) = \ker \delta^1 / \text{im } \delta^0$ and $y \in \text{im } \delta^0$ by construction, so y represents the zero class. (That $y \in \ker \delta^1$ is automatic from $\delta^1 \delta^0 = 0$, but is not even needed: membership in the image is already the conclusion.) $\square$

Claims and non-claims. [REFUTED] The consequence is not subtle. The engine computes its 1-cochain by applying δ to the current state — that is what ω is built from. So the class it then measures in $H^1(\mathcal{C}; \mathcal{F})$ is identically zero, and the growth mechanism that was supposed to be addressed by a nonzero class *has no address*. Every reported $H^1(\mathcal{C}; \mathcal{F})$ diagnostic in that pipeline was measuring the definition of a coboundary, not the state of the system.

The repair requires a source of 1-cochains that is *not* of the form δx . The natural candidate is pairwise judgements formed independently: an organ pair that renders a verdict on their shared edge without either of them consulting a global state. That is new engineering — organs that emit pairwise judgements, storage for 2-simplices, and an actual δ^1 — and it does not exist yet.

One proposed repair in the sources is recorded by the audit as *half circular*, and the circularity must be located precisely before it is adopted. See Open Problem 22.2.

Remark 7.4 (The negative result is publishable). Proposition 7.3 is not merely an internal correction. It identifies a failure mode that any sheaf-based diagnostic over a derived cochain will share, and the derivation is three lines. It belongs in the paper on cohomological obstruction as a stated negative result, not in a footnote.

7.3 The Hodge decomposition

Theorem 7.5 (Hodge decomposition for cellular sheaves). **[PROVED]** Let $L_1 = \delta^0(\delta^0)^\top + (\delta^1)^\top \delta^1$ be the degree-one Laplacian. Then

$$C^1(\mathcal{C}; \mathcal{F}) = \text{im } \delta^0 \oplus \ker L_1 \oplus \text{im } (\delta^1)^\top, \quad (7.1)$$

an orthogonal decomposition, and $\ker L_1 \cong H^1(\mathcal{C}; \mathcal{F})$.

Proof. Orthogonality of $\text{im } \delta^0$ and $\text{im } (\delta^1)^\top$ follows from $\delta^1 \delta^0 = 0$: for $y = \delta^0 a$ and $z = (\delta^1)^\top b$, $\langle y, z \rangle = \langle \delta^0 a, (\delta^1)^\top b \rangle = \langle \delta^1 \delta^0 a, b \rangle = 0$. For $w \in \ker L_1$ we have $0 = \langle L_1 w, w \rangle = \|(\delta^0)^\top w\|^2 + \|\delta^1 w\|^2$, so $w \in \ker(\delta^0)^\top \cap \ker \delta^1$, which is the orthogonal complement of $\text{im } \delta^0 \oplus \text{im } (\delta^1)^\top$. Counting dimensions gives (7.1). Finally each class in $H^1(\mathcal{C}; \mathcal{F}) = \ker \delta^1 / \text{im } \delta^0$ has a unique representative orthogonal to $\text{im } \delta^0$, and that representative lies in $\ker L_1$. $\square$

Intuition. The decomposition sorts every 1-cochain into three kinds of thing: the part that is the disagreement of some global assignment ($\text{im } \delta^0$), the part that is a genuine irreducible contradiction ($\ker L_1$, the harmonic part), and the part that is a local circulation with no bearing on gluing. Proposition 7.3 says that as currently fed, the engine only ever produces cochains of the first kind.

Worked example. On the running example K_4 with the trivial sheaf ($\mathcal{F}(\sigma) = \mathbb{R}$): $\dim C^0(\mathcal{C}; \mathcal{F}) = 4$, $\dim C^1(\mathcal{C}; \mathcal{F}) = 4$ and the complex has one filled triangle so $\dim C^2(\mathcal{C}; \mathcal{F}) = 1$. Since the 1-skeleton is connected, $b_0 = 1$ and $\text{rank } \delta^0 = 4 - 1 = 3$. The only cycle in the 1-skeleton is the triangle $\{S, M, R\}$, and it is filled, so $b_1 = 0$. Consistently: $\dim \ker \delta^1 = \dim C^1(\mathcal{C}; \mathcal{F}) - \text{rank } \delta^1 = 4 - 1 = 3 = \text{rank } \delta^0$, so $H^1(\mathcal{C}; \mathcal{F}) = 0$.

Now delete the 2-simplex, leaving the same 1-skeleton unfilled. Then $C^2(\mathcal{C}; \mathcal{F}) = 0$,

$\ker \delta^1 = C^1(\mathcal{C}; \mathcal{F})$ has dimension 4, and $H^1(\mathcal{C}; \mathcal{F}) \cong \mathbb{R}^{4-3} = \mathbb{R}$, matching $b_1 = 1$: one unfilled loop, one class. This is the sense in which coning (Chapter 15) kills a class — it adds precisely the 2-cells that make the cycle a boundary. $\square$

7.4 The uncertainty stack

Intuition. Coherence is not correctness. Five instruments, ordered by cost, answer five different questions, and they are not interchangeable. The most common error in systems of this kind is to let the cheapest instrument speak for all five.

Instrument	Question	Cost	When
ρ	Are my parts agreeing?	effectively free	every tick
$\hat{\rho}$ vs ρ	Did I <i>know</i> whether I would agree?	free (k -NN)	once history exists
b_0, b_1	What is structurally missing?	medium	on conflict
perturbation	Am I stable to paraphrase?	k model calls	high stakes only
VERIFYOP	Am I actually <i>right</i> ?	medium	promotion gate

Table 7.1: The five instruments and their distinct questions.

Remark 7.6 (Perturbation is a variance, not a derivative). Stability is measured as paraphrase-perturbation variance: run k semantically equivalent paraphrases and measure the spread of outcomes. This replaces an earlier proposal to compute a Fréchet derivative $\|\partial F / \partial x\|$ of the whole pipeline. That derivative does not exist: F factors through discrete token sampling, a DAG generator, and database lookups, none of which is differentiable. Earlier drafts also posited an infinite-dimensional Banach state space for the same purpose; it was removed for the same reason. The replacement costs k model calls and must be rationed.

Remark 7.7 (Expected precision and the feedback edge). $\hat{\rho}$ is a similarity-weighted k -nearest-neighbour prediction of ρ from a log of past (query embedding, ρ) pairs. The *signed* gap $\hat{\rho} - \rho$ separates three states that a single number cannot: *competence* (predicted agreement, observed agreement), *known ignorance* (predicted disagreement, observed disagreement — “confidently, I do not know this”), and *hallucination risk* (predicted agreement, observed disagreement). A VERIFYOP failure updates the store, so self-knowledge improves with use. This is the only instrument in the stack whose accuracy is supposed to increase over time, and it is therefore the one that carries the architectural bet of Chapter 21.

Part VI

Identity

Chapter 8

Identity: What Survives Mutation

NEW IN V2 *This chapter does not exist in the V0 book. It is the centre of V2 and everything after it depends on it.*

Work plan. NEW IN V2

Goal. Make “TATIANA stays itself while it changes” into a number that can be printed and a theorem that can fail. Nothing in this chapter may rest on the word “identity” being intuitively clear.

Raw material. DOCS/TATIANA-V1-IDENTITY-MAP.md §3 (the three levels), §4 (which phase conserves what); MATH-TOOLBOX.md II.2 (the $\mathbb{K}/W$ split and the consolidation formula), III.3 (Construction 6 fixes the growth increment); TATIANA-V2/MATH/phase0_invariant_dimension.md (the first measurement, already run — see step 5).

Steps.

1. **State the split first.** Identity is carried by the crystallized store $\mathbb{K}$, not by the working complex W ; W may be destroyed at the end of every session. Give the reader the reason this is the right place for identity to live *before* any formula: a memory that is destroyed nightly cannot be what makes you you.
2. **Level 0 — the exact invariant.** Define $K(\mathbb{K}) = (\dim \mathcal{F}_{\mathbb{K}}(\sigma))_{\sigma}$, the dimension vector, a K_0 class. Prove that the consolidation rule $R^{\mathbb{K}} \leftarrow \Pi_{O(d)}(R^{\mathbb{K}} + \gamma(\nu)(R^W - R^{\mathbb{K}}))$ conserves it *exactly*. The proof is short and must be given in full: a retraction onto $O(d)$ cannot change a dimension, molding cannot, state relaxation cannot.
3. **The one thing that moves it.** Growth, and only growth, and by a *derived* amount: $\dim \mathcal{F}(w) = \dim H^0(\mathcal{C}; \mathcal{F})(\gamma; \mathcal{F}|_{\gamma})$ from Construction 6 (Chapter 15). Nothing is hand-set. State the theorem shape explicitly, because it is the shape V2 is built to have: *two of three timescales conserve the invariant exactly; the third changes it computably*.
4. **Level 1 — the projective invariant.** Level 0 is coarse: it cannot distinguish two stores with the same dimensions and different maps. State the finer claim, that relative structure is preserved while absolute magnitude is not, $\mathcal{F}_{\mathbb{K}} \sim \lambda \mathcal{F}_{\mathbb{K}}$. Then make the observation that earns it: $\Pi_{O(d)}$ is the polar retraction $R \mapsto UV^T$, which discards scale and keeps direction — written into the design months earlier as a gauge fixing, and only later recognised as the identity-preservation mechanism. Say plainly that the coincidence is the reason to trust it, and that this is **CONTESTED BY AUDIT** until step 5’s measurement is extended to cover it directly.
5. **Report the measurement that exists.** Phase 0 has been run. Reproduce its result here in full, including its controls: on a 3-cycle and on a $b_1 = 0$ path, pre-existing $K(\mathbb{K})$ is untouched across 500 wake/sleep ticks; the single growth event’s size is $k_w = 1$ confirmed by two independent routes (holonomy $\dim \ker(H - \text{Id})$ and cochain-space $\dim \ker(\delta^0(\delta^0)^T)$); $Q(t)$ moves while dimensions stay pinned. Tag **MEASURED**.

6. **State what the measurement does not cover, in the same breath.** Sleep-phase pruning is unimplemented, so the “may decrease” half of the invariant table is untested and the kill-test is *survived, not passed*. No 2-cells; $\gamma(\nu)$ constant; post-growth dynamics unwired; dense rather than Householder maps; synthetic input only. This list is not an appendix item — it goes beside the result.
7. **Level 2 — declare it out of scope and say why.** The full isomorphism class of $\mathcal{F}_{\mathbb{K}}$: a cellular sheaf on a graph is a representation of the incidence quiver, and anything richer than a single cycle is of wild representation type. Krull–Schmidt makes the decomposition into indecomposables unique but not computable. V2 reports levels 0 and 1 and does not chase level 2.

Done when. A reader can state what $K(\mathbb{K})$ is, prove unaided that consolidation conserves it, say exactly how much growth changes it and why that number is not a choice, and list what the existing measurement does not cover.

Work plan. ¡NEW IN V2!

Goal. §: The open question this chapter raises and cannot yet close — whether molding can erode the obstruction that growth depends on.

Raw material. Raised by the Phase 0 experiment; see `TATIANA-V2/MATH/phase0_invariant_dimension.md` §4 and the logbook entry for 2026-08-14.

Steps.

1. State the concern precisely. Molding changes restriction maps; the growth address is a harmonic class determined *by* those maps. In the experiment as run, molding is bounded to one nudge per tick before the same tick’s stall check, so the question never arises. Unbounded, it might.
2. Say why it matters: if slow molding can dissolve a harmonic obstruction, the three-layer handoff of Chapter 9 is not a handoff but a race, and “growth fires when molding stalls” would be a statement about the schedule rather than about the geometry.
3. Do not answer it here. Record it as an open problem with a stated experiment (vary the molding budget per tick; measure whether the harmonic component’s norm decays to below the stall tolerance without growth ever firing) and cross-reference Chapter 22.

Done when. The question is stated sharply enough that someone else could run the experiment that settles it.

Part VII

The Two-Phase Law

Chapter 9

The Two-Phase Law

NEW IN V2 This chapter replaces the role a single global objective would have played. There is no such objective, and the reason is structural.

Work plan. NEW IN V2

Goal. Establish that TATIANA alternates between two phases with different objectives, that this is forced rather than chosen, and that cost lives in exactly one of them.

Raw material. TATIANA_V1_IDENTITY_MAP.md §4 (the table), §4.1 (wake, the Hodge routing law), §4.2 (sleep), §6 (the maintenance cost); MATH_TOOLBOX.md I.5 (predictive coding, $\dot{x} = -Lx$ is sheaf diffusion), II.2 ($\gamma(\nu)$, $\gamma_0 \ll 1$).

Steps.

1. **Open with the table**, since it is the whole chapter in one view: what drives each phase ($q_t \neq 0$ vs $q_t = 0$), what moves (s fast, $\mathcal{F}$ slow, $\mathcal{C}$ grows / $\mathcal{F}$ renormalizes, $\mathcal{C}$ prunes), the objective (discharge discord / pay down maintenance cost), and what happens to $K(\mathbb{K})$ (conserved exactly / may decrease).
2. **Argue both phases are forced.** The model must be able to run with empty input; that requirement alone establishes an offline phase exists. Name it rather than deriving it from biology, and only then note the neuroscience that agrees.
3. **Wake — the Hodge routing law.** On the 2-complex, $C^1(\mathcal{C}; \mathcal{F}) = \text{im } \delta^0 \oplus H^1(\mathcal{C}; \mathcal{F}) \oplus \text{im } (\delta^1)^*$, orthogonally. State and prove the routing claim: *each summand is dischargeable by exactly one layer*. $\text{im } \delta^0$ yields to changing s (the fast flow $\dot{s} = -L_{\mathcal{F}}s + \text{drive}$, provably, since $L_{\mathcal{F}}$ is PSD); $H^1(\mathcal{C}; \mathcal{F})$ does not — it is holonomy of $\mathcal{F}$ around a cycle, and only molding or growth touches it; $\text{im } (\delta^1)^*$ flags a bad 2-cell. Include the Lyapunov argument in full.
4. **The handoff is a stall, not a threshold.** This is the chapter’s sharpest point and must not be blurred. When the constrained descent has converged and discord remains, it is *proven* that no further motion at that level helps. That is the growth trigger, derived from the geometry. Contrast explicitly with a rule of the form $\rho > c$, and say why the latter would be a hand-set number.
5. **Name the one free parameter.** The numerical tolerance on “converged” is an **PROPOSED** engineering choice. State it in the open. Do not hide it inside a λ .
6. **Sleep — renormalize, then down-select.** Apply $\Pi_{O(d)}$ globally: scale drops, relative structure survives (Chapter 8, level 1). Then consult the maintenance cost and prune structure that does not pay for itself.
7. **State where energy lives and why it lives nowhere else.** One term, one phase. Growth adds during wake; pruning removes during sleep; they never bid against each other inside a single tick. Give this its own paragraph — it is the entire answer to the objection that the design needs

a multi-term objective with weights.

8. **The maintenance cost.** $\mathcal{U}(\mathbb{K}) = L(\text{structure}) + L(\text{observations} \mid \text{structure})$. The description-length form is already derived (Chapter 14); what is new here is only its *placement*. Then name the one genuine free parameter of the entire design: the exchange rate between the cost of a deformation and the cost of a new cell, since MDL needs both in the same units. State it in the open and record that if it acquires siblings, the design has regressed.

Done when. A reader can say what moves in each phase, prove that the fast flow cannot discharge a harmonic component, explain why the growth trigger is not a threshold, and name the single exchange rate together with the reason a second one would be evidence of failure.

Work plan. ¡RETIRE!

Goal. §: Record the refused alternative, so it is not proposed again.

Raw material. The predictive-coding discussion in DOCS/Phase I --- Understand the Geometry o.txt (second half); refusal recorded in TATIANA.V1.IDENTITY_MAP.md §8 DON'T 1 and §13.

Steps.

1. State the proposal as it was made and at its strongest: a single cognitive energy functional $F = E_{\text{pred}} + \lambda_C E_C + \lambda_{\mathcal{F}} E_{\mathcal{F}} + \lambda_s E_s + \lambda_I E_I + \lambda_K E_K$, with dynamics choosing whichever admissible transformation most decreases it. It is a coherent design and its appeal should be visible.
2. Give the refusal and its reason: six weights are six hand-set numbers trading against each other every tick, which is exactly the failure mode the discipline rule against hand-set constants exists to prevent. The two-phase separation achieves the same coordination with one exchange rate.
3. Record the standing consequence: *do not add a λ* . A second exchange rate means the two-phase separation has failed and should be re-derived, not patched.
4. Note honestly what the refused proposal got right, since parts of it survive under other names: prediction error precision-weighted is ω (Chapter 5); the compression term is the MDL law (Chapter 14); the stability term is $\gamma_0 \ll 1$'s anti-interference property. What is rejected is the *blending*, not the ingredients.

Done when. Someone rereading the predictive-coding notes can see why they were not adopted wholesale, and which of their parts already live elsewhere in the book.

Part VIII

Dynamics

Chapter 10

Growth Operators and the Algebra



Work plan. **jRECAST**_i

Goal. Keep the typed structural operators. Rebuild the algebra $\mathfrak{D}$ on *measured* relations rather than asserted ones, and be honest that in V2 this chapter is mostly a programme rather than a result.

Raw material. This chapter's existing text; MATH_TOOLBOX.md I.1 (typed ops replacing Pachner moves), I.7 (the curvature controller — derived, unit-tested, never wired); DOCS/Phase I --- Understand the Geometry o.txt (the phased programme: invariants, then primitives, then interaction laws, then the algebra).

Steps.

1. **Keep the typed operators** bind, collapse, split, merge, cone, each logged to a mutation history. Keep the reason they replaced Pachner moves: a knowledge graph is not a manifold, so the preservation guarantee does not transfer. **jKEEP**_i.
2. **Add the invariants step before the primitives.** The V0 text hunts for primitive operations first. The correct order — and the one the source programme itself corrects to — is invariants first: “primitive” is only meaningful relative to what must not move. Each operator must now declare its effect on $K(\mathbb{K})$ (Chapter 8). Make that declaration part of the definition of an operator, not a later audit.
3. **Derive the relations; do not invent them.** The commutators are to be *measured*: apply candidate primitives in both orders on small explicit complexes and compute the difference. State clearly that $\mathfrak{D} = T(V)/I$ is a target, and that I is empty until such measurements exist. An algebra asserted without its relations is bookkeeping.
4. **Report the independence-relation test honestly.** If the independence relation is empty, $\mathfrak{D}$ is the free algebra and the whole apparatus counts directed acyclic graphs. Record this as the falsifier for this chapter, and record that it has not been run for V2's operator set.
5. **Keep the curvature controller with its warning.** “Run the flow, skip the surgery” is right, and Forman's weights are *not* free — they are fixed by whichever Laplacian the system actually prints ω and ρ from, so using a Hebbian coupling as the Forman weight computes the curvature of a *different* operator: a consistent-looking number about the wrong thing. Keep the flagged prediction that a low-precision edge may drift positive and be folded *because we are unsure of it*, which is backwards; check the sign behaviour before trusting the controller.
6. **Keep the floor derivation as a template.** The exact multiplicative flow can never sever an edge; what severs is the forward-Euler discretisation flipping sign — so the old weight floor was patching a numerical artefact that had been misdiagnosed as a safety mechanism. The replacement is exponential integration plus a control barrier, which turns out to be exactly the

object neuroscience calls weight-dependent plasticity. Keep also the floating-point trap: bare exponential integration underflows to zero and severs silently; only the barrier survives floating point.

7. **State the implementation status plainly.** No line of the curvature controller has ever moved a real edge weight. **[PROPOSED]**.

Done when. Every operator declares its effect on the invariant, and the chapter distinguishes what is proved from what is a programme awaiting measurement.

Intuition. Learning *is* change of shape. The complex binds organs that work together, prunes idle ones, and — when a procedure repeatedly succeeds — promotes it to a first-class operator. Four axes of growth, one of which grows the very set of moves the system can make.

10.1 The structural operators

Definition 10.1 (Base moves). The base moves on $\mathcal{C}$ are the typed operators

$$\{\text{bind, collapse, split, merge}\},$$

each logged to an immutable mutation history.

Remark 10.2 (“Pachner move” was discarded, correctly). Earlier drafts called these Pachner moves. A knowledge complex is not a manifold, so the invariance guarantees that make Pachner moves useful — that any two triangulations of a PL manifold are connected by a finite sequence of them — simply do not transfer. Borrowing the name would have imported a theorem that does not apply.

Construction 10.3 (The four growth axes). Evolution proceeds along

$$\underbrace{\delta\mathcal{C}_{\text{fine}}}_{\text{concepts}}, \quad \underbrace{\delta\mathcal{C}_{\text{coarse}}}_{\text{coalitions}}, \quad \underbrace{\delta\mathcal{F}}_{\text{stalk and restriction data}}, \quad \underbrace{\delta\mathfrak{D}}_{\text{new operators}}.$$

The fourth axis is the essential one: the operator algebra is time-dependent, $\mathfrak{D}(t) \rightarrow \mathfrak{D}(t+1)$.

Definition 10.4 (Hebbian coupling and decay). For a coalition σ with weight $w(\sigma, t) \in [0, 1]$: co-activation increments w ; idle steps apply $w(\sigma, t+1) = (1-\gamma)w(\sigma, t)$ for a decay rate $\gamma \in (0, 1)$; a coalition falling below a bind threshold *collapses*. A pair past the bind threshold is a 1-simplex of K .

Proposition 10.5 (Idle coalitions collapse in finite time). **[PROVED]** *If σ is idle from time t_0 and the bind threshold is $\theta > 0$, then σ collapses at the first $t \geq t_0 + \log_{1/(1-\gamma)}(w(\sigma, t_0)/\theta)$.*

Proof. Iterating the decay gives $w(\sigma, t_0 + m) = (1-\gamma)^m w(\sigma, t_0)$, which falls below θ as soon as $m \ln(1-\gamma) < \ln(\theta/w(\sigma, t_0))$; since $\ln(1-\gamma) < 0$ this is $m > \log_{1/(1-\gamma)}(w(\sigma, t_0)/\theta)$. $\square$

Construction 10.6 (Consolidation and microneurons). Over the weight filtration, persistent homology reads which coalitions and concept-cycles *survive*; survivors are promoted into permanent structure. A *microneuron* is a promoted sub-tree: when a composite DAG repeatedly resolves conflict *and* passes verification, it is named and thereafter emitted by the planner as a single atomic generator of $\mathfrak{D}$. This is $\delta\mathfrak{D}$ made concrete. The promotion gate is **VERIFYOP** (Chapter 17): a composite becomes a generator only if it survives the external oracle.

Claims and non-claims. The requirement that promotion pass VERIFYOP is doing real work. Without it, “repeatedly resolved conflict” promotes whatever procedure most reliably makes the organs agree — which, in a system whose only cheap signal is agreement, selects for procedures that manufacture consensus rather than truth. Coherence is a control signal, not a fitness function, and the gate is what keeps that distinction enforced.

Chapter 11

The Fine Dynamics

Intuition. Inside an organ, reasoning is a *flow*. New material perturbs the internal state and a gradient descent pulls it back toward internal coherence, subject to hard constraints (axioms that must hold) and soft constraints (inferred facts, bendable).

Construction 11.1 (Fine state and deductive flow). The fine state of an organ is a configuration $x \in (\mathbb{R}^d)^m$ over its m active concepts, with a scalar *conflict functional* $\Phi(x) \geq 0$ measuring internal inconsistency — the fine-level analogue of ω . Reasoning integrates

$$\dot{x} = -\nabla\Phi(x) \tag{11.1}$$

by a fixed-step fourth-order Runge–Kutta scheme. The reported δ -strain of a step is the achieved change $\Delta\Phi$. Two constraint types shape Φ : *rigid* constraints (user-stated axioms) act as hard orthogonal projections onto an affine subspace via a projector P ; *fluid* constraints (inferred facts) act as soft quadratic penalties.

Proposition 11.2 (The flow is a descent). **[PROVED]** *Along any solution of (11.1), $\frac{d}{dt}\Phi(x(t)) = -\|\nabla\Phi(x(t))\|^2 \leq 0$, with equality exactly at critical points. If Φ is bounded below — which it is, by $\Phi \geq 0$ — then $\Phi(x(t))$ converges and $\nabla\Phi(x(t)) \rightarrow 0$ along a subsequence.*

Proof. Chain rule: $\frac{d}{dt}\Phi = \langle \nabla\Phi, \dot{x} \rangle = -\langle \nabla\Phi, \nabla\Phi \rangle$. Monotone and bounded below implies convergent; integrating $\int_0^\infty \|\nabla\Phi\|^2 dt = \Phi(x(0)) - \lim \Phi < \infty$ forces $\liminf \|\nabla\Phi\| = 0$. $\square$

Remark 11.3 (Observed sign behaviour). On a live run, injecting a new concept raised Φ by a structural jump (δ -strain = +0.202) and the subsequent flow pulled it back down (δ -strain = −0.081): a structural expansion followed by a deductive relaxation, which is the intended dynamics. Note that the expansion is *not* a descent step — it is an exogenous change of the functional itself, and only the relaxation is governed by Proposition 11.2.

Claims and non-claims. The closed form of Φ is an implementation-level choice — a penalised quadratic on the fine complex — and is deliberately not over-specified here. What is mathematically committed is (11.1) (a gradient flow on a nonnegative conflict functional) and the rigid/fluid split as hard projection versus soft penalty. Convergence to a *global* minimum is not claimed and would be false for a non-convex Φ .

Chapter 12

Scheduling: Antichains, and Why “Operad” Is Not Earned

Intuition. A reasoning act is a small program: a directed acyclic graph of operators, some depending on others. Independent operators can run at once, and the scheduler slices the DAG into layers of mutually independent work.

Definition 12.1 (Antichain foliation). A reasoning act is a DAG $T = (N, \prec)$ of operator nodes. The scheduler foliates T into the antichain layers of $\prec$: layer 0 is the set of minimal nodes, and layer $\ell + 1$ is the set of nodes all of whose predecessors lie in layers $\leq \ell$. Each layer is executed in parallel. Correctness follows because nodes in one antichain share no $\prec$ -dependency. Node *support* sets index which fine-complex vertices an operator reads and writes, so slices with disjoint support are safe to co-execute.

Remark 12.2 (The intended structure is a trace monoid). What Definition 12.1 describes is antichain layering of a DAG. An operad requires multi-input composition together with equivariance, associativity and unit axioms, none of which are stated or used anywhere. The structure the design is reaching for is a *trace monoid* $\mathbb{M}(\Sigma, I)$ in the sense of Mazurkiewicz: the free monoid on the operator alphabet modulo $ab = ba$ for pairs in an *independence relation* I , whose canonical Foata normal form is exactly the antichain layering the scheduler computes. That identification would be a *sharpening* rather than a downgrade, since it supplies normal forms, a Hilbert series, and a canonical form for deduplication. The terminology should be corrected regardless of whether the identification is completed.

Claims and non-claims. [REFUTED] The trace-monoid identification **does not currently hold**, and the reason is three separate defects, measured 2026-07-29 against the deployed operator set.

(i) *The relation is not symmetric.* In the scheduler, admitting *any* empty-support node — including a read-only one — marks the whole slice as a global mutation and blocks every later node. So 8 of the 15 operator pairs can share a slice in one order and not in the other. An independence relation is symmetric by definition; a non-symmetric relation defines no trace monoid, hence no Foata normal form and no Hilbert series. This is very likely an unintended scheduling behaviour and is worth fixing on its own merits.

(ii) *The support sets are placeholders.* Every support accessor returns a hardcoded

literal — $\emptyset$, $\{0\}$ or $\{1\}$ — with three of the six operators returning the same $\{0\}$, and a comment standing in for an analysis of what the operator actually touches. Any invariant computed from these describes six constants, not the system.

(iii) *The convention for $\emptyset$ is inverted.* In the engine an empty support means GLOBAL: touches everything, must run isolated. Read as a set, $\emptyset$ is disjoint from everything and would mean independent of everything. The two readings give capacities differing by a factor of 1.8 ($\kappa = 3.00$ versus $\kappa = 5.45$, against $\kappa = 6$ for the free monoid), so the quantity is not well defined until the convention is fixed.

Under the reading the engine actually implements, the algebra is *nearly free* — about 9% below the free monoid — because three of six operators share one support. The order of repair is: fix the symmetry defect; derive real read/write sets, including the shared state the current model ignores entirely (the store, the obstruction counter Ω , the mutation log, and the coarse stalks that π_v overwrites after every execution); only then recompute I . Operators that all mutate Ω do not commute whatever their vertex supports say. Until that is done, no capacity, Koszulity, or resolution claim about $\mathfrak{D}$ should appear anywhere.

Chapter 13

The Lift–Project Adjunction

Work plan. **¡KEEP!**

Goal. Keep the adjunction — it is standard, genuinely proved, and now does double duty as the memory architecture and as the candidate reproduction morphism.

Raw material. This chapter’s existing text; MATH_TOOLBOX.md II.2 (the $\mathbb{K}/W$ two-complex model, the consolidation rule, the measured $Q(t)$ trajectory), II.1 (what was tried and abandoned for memory identity first).

Steps.

1. **Keep the theorem.** For a full subposet inclusion ι : ι^* restricts (instantiate, store $\rightarrow$ cache), $\iota_!$ is extension by zero (write back only what was touched), ι_* fills untouched cells by limits, and the (co)units are isomorphisms so $\iota^*\iota_! \cong \text{Id}$. **[PROVED]**, standard.
2. **State the write-back choice and its reason.** $\iota_!$, never ι_* : the cache must assert *nothing* about untouched cells, which is the sheaf-theoretic form of “no measurement is not a measurement of zero.”
3. **Present consolidation as the anti-interference mechanism.** $\gamma_0 \ll 1$ is what lets new learning coexist with old memory without overwriting it, and the session’s state s_W is *never* written back — content and wiring persist, the episode does not.
4. **Report that $\gamma_0 \ll 1$ was forced twice, independently.** Once by the anti-interference argument, and once by measurement: no restriction-map parameterisation is affordable to learn from a single tick (the cheapest is over-parameterised threefold, a full orthogonal map by over five hundred). Two independent routes to one conclusion is worth stating as such.
5. **Keep $Q(t)$ as an instrument.** Mean $\|R - \text{Id}\|_F^2$; constant sheaf $\iff Q = 0$; measured moving monotonically off zero across four ticks, and *exactly* zero when nothing is verified. “ $Q(t)$ leaving zero is not a proxy for consolidation; it is the definition.” **[MEASURED]**.
6. **Keep the two bugs the tests caught**, since both would have rotted silently: a constructor that pre-filled every edge with identity so learned maps were dropped on rebuild, and a single Householder reflection landing in the wrong component of $O(d)$ so consolidation correctly refused to interpolate and $Q(t)$ would have stayed zero with every test passing.
7. **Add the forward reference to reproduction.** This same adjoint pair is Chapter 19’s candidate morphism. Note it here.
8. **Keep the abandoned alternatives with their reasons.** Formal Concept Analysis **[REFUTED]** as substrate (data-determined; its canonicity comes precisely from discarding the path-dependence that makes a memory a memory — memory must *grow*, not be recomputed), and the K_0 /Jordan–Hölder schema **[REFUTED]** (both candidate categories fail for opposite reasons). Both were dissolved by the growth-history identity this chapter provides.

Done when. The adjunction is proved, the write-back choice is justified by the $\perp$ -invariant, and its second role in reproduction is signposted.

Intuition. Two worlds: free-form natural language, and structured reasoning DAGs. One map *lifts* language into a validated DAG; a partner map *projects* a DAG back into language. Meaning that carries no reasoning content collapses under the lift — it lands in the kernel.

Construction 13.1 (The boundary morphisms). Let $\mathcal{L}$ send a natural-language request to a validated reasoning DAG, and $\mathcal{M}$ send a DAG to a natural-language rendering. The lift is guarded by a validator rejecting malformed DAGs at the boundary: duplicate identifiers, dangling children, cycles by depth-first colouring, absence of a terminal output node, and disconnection. The downstream engine therefore only ever receives well-formed programs.

Remark 13.2 (The kernel behaves as intended). Semantically void input maps into $\ker \mathcal{L}$: a joke embedded in a mathematical prompt produced zero DAG nodes, exactly as a kernel element should. This is an observation on one input, not a theorem.

Claims and non-claims. $\mathcal{L} \dashv \mathcal{M}$ is the intended categorical *framing* — lift left adjoint to project — and it organises the design usefully. The unit and counit are not constructed here and the triangle identities are not verified, so “adjunction” should be read as a structuring principle plus two concrete morphisms with an observed kernel behaviour, and not as a proved theorem. Stating this is part of the honesty invariant rather than an apology for it.

Part IX

The Later Constructions

Chapter 14

The MDL Growth Law

Work plan. ¡RECAST!

Goal. Keep Construction 7 and give it its V2 placement: this is the *maintenance cost* of the sleep phase, and it is the only place cost lives in the entire architecture.

Raw material. This chapter’s existing text; MATH_TOOLBOX.md III.4 (Construction 7, the per-direction code, the measurement); DOCS/DERIVATION_MDL_GROWTH_LAW.md; TATIANA_V1_IDENTITY_MAP.md §6 (placement, and the one free parameter).

Steps.

1. **Keep the superseded law as a worked failure.** The original had a numerator overcounting by a factor of k (fixed by holonomy propagation — only the first leg needs describing, the rest are determined) and, fatally, a denominator that did not depend on concept size at all, making a zero-dimensional concept always optimal: “attach always, with a concept that carries nothing.” Keep this; it shows exactly how a plausible law can invert its own purpose.
2. **State Construction 7’s repair.** A per-direction code: fix a direction in the limit, take its k readings around a traversal, and ask whether they agree. The saving per traversal follows from the intraclass correlation, and the law becomes a threshold on recurrence count that is never met when the saving is non-positive.
3. **Report the measurements.** Within 0.042 bits/traversal of exact quantised-Gaussian entropy; over-prices its own reference code so the threshold is *conservative* — it under-attaches, never over-attaches; and a shuffled-readings control drops the saving from +5.289 to −0.437 bits, so the saving measures the recurrence and not the coder. [MEASURED].
4. **Keep the restored consequences.** Size-dependence returns per-direction; the subspace-selection rule (rank by reliability, keep a prefix) is now a *theorem* rather than a proposal; and a reliability floor exists independent of frequency.
5. **Keep the open self-reference problem.** Taking bits-per-real as $\frac{1}{2} \log n$ is self-referential but resolvable by iteration — *except* that it degenerates exactly where the growth law lives: at $n = 1$ the model is free and over-generation returns. Do not adopt it as derived-therefore-safe. State the alternatives.
6. **Keep the stated narrowing.** The law prices what was there, not which succession fired — a strict under-estimate, making it conservative by construction rather than by luck.
7. **Add the V2 placement paragraph.** This cost is consulted during *sleep* only. It never bids against growth inside a single tick. Name the exchange rate between deformation cost and new-cell cost as the one genuine free parameter of the design, and record that acquiring a second one is evidence the two-phase separation has failed.

Done when. The law is stated with its measurements and its open self-reference problem, and its placement in the sleep phase is explicit.

The question this chapter answers: *when is a new concept worth inventing?* Earlier designs answered it with a threshold someone chose. This one derives a threshold instead.

14.1 Two-part description length

Intuition. A memory is a compression of experience. Minimum description length makes that literal: the total cost of an explanation is what it takes to write down the model plus what it takes to write down the data *given* the model. Adding a concept always makes the first term worse. It is worth doing exactly when it makes the second term better by more.

Definition 14.1 (Two-part MDL).

$$L_{\text{total}} = \underbrace{L(\text{model})}_{\text{the memory}} + \underbrace{L(\text{data} \mid \text{model})}_{\text{the experience}}.$$

The *model* is the complex $\mathcal{C}$ together with the sheaf $\mathcal{F}$ — what the system *is*. The *data* is the observed record: the sequence of assemblies and the succession relation read off it — what the system has *seen*.

Neither term alone is the criterion. Minimising the model term alone forbids all growth; minimising the data term alone is achieved by memorising everything. The criterion is the sum.

14.2 The model term

Proposition 14.2 (Cost of coning a cycle). **[PROVED]** *Let γ be a 1-cycle of length k , and cone it with a new vertex w carrying a stalk of dimension d_w . Writing b for bits per real number, the added model cost is*

$$\Delta L(\text{model}) = b d_w \left(1 + \sum_{e \in \gamma} d_e \right). \quad (14.1)$$

Proof. Three kinds of new object are described. The vertex with its stalk costs $d_w \cdot b$ bits. Each of the k restriction maps out of $\mathcal{F}(w)$ is a $d_e \times d_w$ matrix, costing $d_w d_e b$ bits, summed over $e \in \gamma$. The k new triangles cost *nothing*: they are determined by the cone construction, not described separately. Summing gives (14.1). $\square$

Remark 14.3 (Two consequences, both load-bearing). First, the cost is *linear in d_w* . The dimension of the new stalk is the dominant free quantity, so whatever fixes d_w fixes most of the cost — which is what Section 15.6 is for. Second, *triangles are free*. They are implied by the cone, so the k new 2-cells cost nothing to describe. This is a real asymmetry, not a bookkeeping convenience, and it is why coning is cheap relative to what it buys.

14.3 The data term and the law

Proposition 14.4 (Data-term saving). **[PROVED]** *Let the cycle γ occur n times in the record. Without the concept, each traversal is coded as k separate transitions, each costing its full surprisal; with the concept it is a single named event. Then*

$$\Delta L(\text{data} \mid \text{model}) = -n(c_{\text{old}} - c_{\text{new}}), \quad (14.2)$$

where c_{old} is the cost of coding one traversal the long way and c_{new} the cost of naming it.

Theorem 14.5 (The growth law). **[PROVED]** *Attach the concept if and only if $\Delta L_{total} < 0$, that is*

$$n > \frac{b d_w (1 + \sum_{e \in \gamma} d_e)}{c_{old} - c_{new}} \quad (14.3)$$

Proof. Add (14.1) and (14.2) and require the sum negative; rearrange, using $c_{old} > c_{new}$. $\square$

Intuition. Read what this is: a *recurrence threshold that is computed, not chosen*. It is the ratio of what a concept costs to store against what it saves per use — the number of times a pattern must repeat before it is worth naming.

Three consequences, two of which were not asked for:

1. n is a number the engine can print, rather than a parameter a human sets.
2. **It orders the holes.** When several harmonic classes compete, fill them in decreasing $|\Delta L_{total}|$. This removes a decision that had been silently delegated to a human.
3. **It predicts non-attachment.** A one-off inconsistency is *noise to be tolerated*, not a concept to be invented. The earlier threshold-based law had no way to express this, which is exactly how it would have over-generated.

Chapter 15

The Coned Concept

Work plan. **¡KEEP!**

Goal. Keep Construction 6 essentially as written — it is the most-verified construction in the record — and add the one upgrade V2 needs: its output dimension is what moves the invariant, so this chapter and Chapter 8 must agree exactly.

Raw material. This chapter’s existing text; MATH_TOOLBOX.md III.3 (derivation, two self-corrections, the measurements, provenance); TATIANA-V2/MATH/phase0_invariant_dimension.md (the dimension confirmed two independent ways).

Steps.

1. **Keep the derivation and its punchline.** Consistency on the new edges forces the value at w to *generate* the section over the whole cycle, and substituting into the old edges gives verbatim the categorical cone condition. The cone condition was not imposed — it fell out of demanding the topological cone do its job. **[PROVED]**.
2. **Keep the identification.** $\mathcal{F}(w) := \lim D_\gamma \cong H^0(\mathcal{C}; \mathcal{F})(\gamma; \mathcal{F}|_\gamma)$: the new concept is *not* a summary of the cycle’s members but their *compatible part*. This sentence is the chapter.
3. **Keep both self-corrections visible.** First, an earlier draft argued the limit is “the smallest object admitting the required maps” — backwards; every cone factors uniquely through the limit, so it is the *largest* non-redundant choice, a ceiling and not a floor. Second, the “refusal branch” (nothing consistent, so the construction declines) is logically real but *structurally unreachable*: a cycle carries harmonic mass if and only if it has sections.
4. **Keep the upgrade from preference to necessity.** The coned complex is a cochain complex *if and only if* $\mathcal{F}(w)$ is a cone over the diagram. Construction 6 is therefore not the best choice among several — it is the only choice for which the coned object is a sheaf at all. The universal property was never a preference; it was the existence condition. **[PROVED]**, and measured: an arbitrary $\mathcal{F}(w)$ gives 1.756, the limit gives 4.4×10^{-16} .
5. **Keep the parity observation.** Reachability of the degenerate case is governed by the parity of d , since every element of $SO(\text{odd})$ fixes an axis. This is a fact about this architecture specifically, because its restriction maps are orthogonal by construction.
6. **Keep the provenance paragraph, including the direction.** The universal-property route to a new concept is Goguen’s, not this project’s — and the variance is *opposite*: his blends are colimits, a sheaf’s restriction maps force a limit. “Copying him would have gotten it backwards.” What is genuinely original is that the sheaf’s own restriction maps supply the diagram, so nothing new is introduced beyond what the memory already held.
7. **Add the invariant link.** $\dim \mathcal{F}(w) = \dim H^0(\mathcal{C}; \mathcal{F})(\gamma; \mathcal{F}|_\gamma)$ is exactly the amount by which $K(\mathbb{K})$ changes, known *before* the growth decision, from the same computation that produced the address. Cross-reference Chapter 8 and report that both routes to this number have been

computed and agree. **[MEASURED]**.

8. **Record what is still not derivable.** The label for the new concept is not derivable from a diagram. Name this as the one place genuine novelty enters rather than being computed, and connect it to Chapter 20.

Done when. The construction is proved to be forced rather than chosen, and its output dimension is stated identically here and in Chapter 8.

This is the central construction of the later work: the stalk of a newly-grown concept, *derived* rather than chosen.

15.1 The topological cone

Construction 15.1 (Coning a cycle). Let $\gamma = (v_1, e_1, v_2, \dots, v_k, e_k, v_1)$ be a 1-cycle carrying harmonic mass, with $e_i = \{v_i, v_{i+1}\}$ (indices mod k). Attach a vertex w , edges $f_i = \{w, v_i\}$, and triangles $t_i = \{w, v_i, v_{i+1}\}$.

Proposition 15.2 (The cone kills the cycle). **[PROVED]** With $\partial t_i = e_i - f_{i+1} + f_i$,

$$\partial\left(\sum_i t_i\right) = \sum_i e_i + \sum_i (f_i - f_{i+1}) = \gamma,$$

since the f terms telescope to zero. Hence γ becomes a boundary.

15.2 The stalks, chosen minimally

Construction 15.3 (New edge stalks). Set

$$\mathcal{F}(f_i) := \mathcal{F}(v_i), \quad \mathcal{F}_{v_i \trianglelefteq f_i} := \text{Id}.$$

Claims and non-claims. This is an *engineering choice*, and a deliberately empty one. Any distortion introduced here would be a free parameter, and whatever the construction then achieved could be attributed to a cleverly chosen edge stalk rather than to the concept. All the content is forced into $\mathcal{F}(w)$ and the maps out of it, which is where it can be reasoned about.

The only remaining unknowns are $\mathcal{F}(w)$ and the legs $p_i := \mathcal{F}_{w \trianglelefteq f_i} : \mathcal{F}(w) \rightarrow \mathcal{F}(v_i)$.

15.3 The cone condition is forced, not imposed

Proposition 15.4 (Derivation of the cone condition). **[PROVED]** Write $r_i^- := \mathcal{F}_{v_i \trianglelefteq e_i}$ and $r_i^+ := \mathcal{F}_{v_{i+1} \trianglelefteq e_i}$. For a 0-cochain x :

- (i) δx vanishes on the new edges if and only if $x_{v_i} = p_i(x_w)$ for every i ;
- (ii) given (i), δx vanishes on the old edges for every $x_w \in \mathcal{F}(w)$ if and only if

$$\boxed{r_i^+ \circ p_{i+1} = r_i^- \circ p_i \quad \text{for all } i} \tag{15.1}$$

Proof. (i) On the new edge f_i , using Construction 15.3,

$$(\delta x)_{f_i} = \mathcal{F}_{w \trianglelefteq f_i} x_w - \mathcal{F}_{v_i \trianglelefteq f_i} x_{v_i} = p_i(x_w) - x_{v_i},$$

which vanishes exactly when $x_{v_i} = p_i(x_w)$.

(ii) Substituting $x_{v_i} = p_i(x_w)$ into the old edge e_i ,

$$(\delta x)_{e_i} = r_i^+ p_{i+1}(x_w) - r_i^- p_i(x_w) = (r_i^+ p_{i+1} - r_i^- p_i)(x_w).$$

A linear map vanishes on all of $\mathcal{F}(w)$ if and only if it is the zero map, which is (15.1). $\square$

Intuition. Part (i) says something worth pausing on: **the value at w generates the section over the whole cycle**. That is what it means for a concept to explain a loop — the loop stops being k independent facts and becomes one.

Condition (15.1) is verbatim the definition of a *cone* over the diagram

$$D_\gamma : \quad \mathcal{F}(v_1) \xrightarrow{r_1^-} \mathcal{F}(e_1) \xleftarrow{r_1^+} \mathcal{F}(v_2) \xrightarrow{r_2^-} \mathcal{F}(e_2) \leftarrow \dots$$

with apex $\mathcal{F}(w)$ and legs p_i . The categorical cone condition was not imposed on the construction; it fell out of demanding that the topological cone do its job.

15.4 The construction

Construction 15.5 (The coned concept).

$$\boxed{\mathcal{F}(w) := \varprojlim D_\gamma, \quad p_i := \text{the limit's projections.}}$$

Proposition 15.6 (Concrete description). **[PROVED]** For finite-dimensional stalks,

$$\varprojlim D_\gamma = \left\{ (x_1, \dots, x_k) \in \bigoplus_i \mathcal{F}(v_i) \mid r_i^- x_i = r_i^+ x_{i+1} \ \forall i \right\} \cong H^0(\gamma; \mathcal{F}|_\gamma).$$

Proof. The limit of a finite diagram of vector spaces is the subspace of the product on which all the diagram's maps agree, which is the displayed equaliser. Comparing with Definition 4.1 restricted to the subcomplex γ , the agreement conditions are exactly $\delta x = 0$, so the space is $\ker \delta|_\gamma = H^0(\gamma; \mathcal{F}|_\gamma)$. $\square$

Intuition. In words: the new concept *is* everything that was consistent around the loop but could not be glued. It does not summarise the cycle's members; it *is* the compatible part of them.

Three properties, each of which had been written down separately as a requirement:

Property	Why it matters
canonical — unique up to unique isomorphism	no hand-made choice enters the construction
path-dependent — depends on D_γ , i.e. on which cycle history produced	precisely the property whose absence caused an earlier formal-concept-analysis approach to be rejected
degenerate case	if $H^0(\gamma; \mathcal{F} _\gamma) = 0$ the limit is the zero space — but see Proposition 15.7

15.5 The refusal branch is unreachable

Proposition 15.7 (Harmonic mass implies sections). **[PROVED]** *On a k -cycle with vertex and edge stalks all of dimension n , and with no 2-cells,*

$$\dim H^1(\gamma; \mathcal{F}|_\gamma) = \dim H^0(\gamma; \mathcal{F}|_\gamma).$$

Consequently the cycle carries harmonic mass if and only if it has nonzero sections.

Proof. For a 1-dimensional complex the Euler characteristic satisfies $\chi = \dim C^0(\mathcal{C}; \mathcal{F}) - \dim C^1(\mathcal{C}; \mathcal{F}) = \dim H^0 - \dim H^1$. On a k -cycle with all stalks of dimension n there are k vertices and k edges, so $\dim C^0(\mathcal{C}; \mathcal{F}) = \dim C^1(\mathcal{C}; \mathcal{F}) = kn$ and $\chi = 0$, giving the equality. With no 2-cells the harmonic space is H^1 . $\square$

Claims and non-claims. An earlier draft claimed the degenerate case is meaningful — “nothing was consistent around that loop, so the construction refuses rather than inventing”. True, and it never happens. By Proposition 15.7, $H^0 = 0$ implies no harmonic mass, which implies no growth address, which implies the growth law never fires on that cycle at all. Confirmed numerically: the degenerate case reports that the harmonic space is already empty before coning, so there is nothing to kill.

In general the gap is $\dim H^1 - \dim H^0 = k(n_e - n_v)$, so the branch *could* become reachable if edge stalks are ever given a different dimension from vertex stalks. They are not, currently.

15.6 The limit is a ceiling, not a floor

Claims and non-claims. [REFUTED] An earlier draft justified Construction 15.5 by asserting that the limit is “the smallest object admitting the required maps”. That is backwards, and the error is worth keeping visible.

Proposition 15.8 (Universality bounds from above). **[PROVED]** *Every cone over D_γ factors uniquely through the limit. If a candidate apex $\mathcal{F}(w)'$ has jointly injective legs — i.e. carries no data that no restriction map ever sees — then that factoring map is injective, so*

$$\dim \mathcal{F}(w)' \leq \dim \varprojlim D_\gamma.$$

Proof. The universal property of the limit gives a unique $\phi : \mathcal{F}(w)' \rightarrow \varprojlim D_\gamma$ commuting with the legs. If $\phi(z) = 0$ then every leg $p'_i(z) = 0$; joint injectivity of the legs then forces $z = 0$, so ϕ is injective and the dimension inequality follows. $\square$

So the limit is the *largest non-redundant* choice, not the smallest — a ceiling on what a new concept can usefully carry. The two determinations are therefore genuinely separate, and both are derived:

1. **Consistency fixes the shape.** $\mathcal{F}(w)$ must be a cone over D_γ ; the limit is the universal one.
2. **MDL fixes the size.** By (14.1) the cost is linear in d_w , so the optimum is a *subspace* of the limit: keep the directions that pay for themselves, discard the rest.

Chapter 16

Retrieval: Rank, Do Not Threshold

Work plan. **RECAST**

Goal. Keep the empirical result, which is large and well-measured, but change what the chapter is for: V2's own diagnosis is that retrieval cannot live in a metric at all, so this chapter becomes the evidence for that conclusion rather than a description of a component.

Raw material. This chapter's existing text; MATH_TOOLBOX.md V.1–V.2 (the bug, the fix, the self-audit), VII.1 (the six measurements that together say retrieval needs a generative process).

Steps.

1. **Keep the bug and its magnitude.** A relevance threshold compared against squared Bures–Wasserstein distance, with an unbounded scan and no limit, admitted 0.31% of ground-truth dependencies; at the shipped default, literally zero of three thousand ticks retrieved a pair. **[MEASURED]**.
2. **Keep all three root causes**, since each is instructive: it was a near-duplicate filter rather than a relevance filter; embeddings sit on a large common-mode pedestal that any absolute threshold mostly measures; and the epistemic term is dimension-extensive while the semantic term is dimension-intensive, so summing them lets the former dominate by two orders of magnitude regardless of meaning.
3. **Keep the fix and the honest framing of its size.** Rank, do not threshold; rank on cosine, not squared distance (stored means are not unit-norm, so squared-distance ranking silently penalises general concepts). Worth 0.31% → 46.6%. State the lesson explicitly: the whole design argument was about a four-point effect while the actual fix was worth 140×.
4. **Keep the self-audit.** Two claims about the fix were themselves overstated and walked back — an optimum read off a flat sweep, and a parameter called “derived” that was an unmeasured budget coincidence. “Two independent constraints landing on the same number is exactly the tidiness motivated reasoning produces.” Keep this sentence.
5. **Keep the refuted alternative.** Diffusion/heat-kernel retrieval, twelve configurations, none beating raw cosine, degrading monotonically with diffusion time — the pedestal makes the walk near-uniform so it amplifies hubness rather than removing it. **[REFUTED]**.
6. **Add the V2 recast.** State the conclusion this chapter is now evidence for: retrieval cannot live in the metric alone, cannot live in a single relation (citation and embedding similarity agree under half the time), and cannot live in the corpus. What is missing is a process that decides what to retrieve *about* — which in V2 is the wake phase's drive, not a lookup. Reframe accordingly and forward-reference Chapter 9.
7. **Carry the standing warning.** A simulator produces the topology its rules imply. Moving from “assume a structure and test for it” to “generate the structure” is a stronger design and *also* a way to hide assumptions inside an update rule. No claim about emergent structure without a

validation battery built *before* the generator.

Done when. The chapter reads as the argument that retrieval needs a process rather than a metric, with the measurements as evidence rather than as the point.

Intuition. Retrieval decides which stored concepts an organ sees. A threshold on distance answers “which concepts are close enough?”, which sounds right and is not: it makes the number of results depend on how the embedding space happens to be scaled, and it silently returns nothing at all when the query lands in a sparse region.

Construction 16.1 (Bounded rank query). Replace the ε -ball query with a bounded rank query returning the top k stored concepts, ranked by *cosine similarity* to the query mean, using a bounded max-heap of size k . Zero-norm vectors are dropped rather than scored, since a zero vector has no direction.

Proposition 16.2 (Why cosine and not squared distance). **[PROVED]** *Expanding,*

$$\|\mu_q - \mu_i\|^2 = \|\mu_q\|^2 - 2\langle \mu_q, \mu_i \rangle + \|\mu_i\|^2.$$

For a fixed query, $\|\mu_q\|^2$ is constant and drops out of any ranking. The term $\|\mu_i\|^2$ does not. Hence squared distance reproduces the cosine ranking if and only if every stored mean is unit-norm.

Claims and non-claims. Stored means are *not* unit-norm. By Construction 6.2, π_v is a weighted centroid of unit vectors, so its norm lies below 1 and falls further the more spread the concepts it summarises. Ranking by squared distance would therefore apply a per-concept penalty *proportional to how broad a concept is*, pushing exactly the general concepts down the list for a reason unrelated to the query.

There is also an evidential argument, which is decisive on its own: the measured recall figure against which the retrieval fix is accepted was obtained on a *cosine* ranking. Ranking by squared distance is a different ordering carrying an unmeasured bias, which would have made the acceptance criterion meaningless.

Remark 16.3 (Cost). Scoring needs only μ ; the low-rank factor U , the floor D and the stored reasoning text are not needed unless the row enters the heap. Reading μ , scoring, and deserialising the rest only on entry drops the per-row cost from $O(dk^2 + k^3)$ to $O(d)$ for the large majority of rows.

Claims and non-claims. k is an *engineering choice* ($k = 24$), not a derived quantity. What changed is that it is now constrained on both sides — too small starves the organ, too large defeats the point of retrieval — rather than being an unexamined constant. Saying so is the difference between a parameter and a magic number.

Part X

Verification and Control

Chapter 17

Verification: the External Oracle

Work plan. **¡RECAST!**

Goal. Keep the trichotomy, which is a real epistemic fix. Retitle and re-found the chapter: in V2 there is no “external oracle” to appeal to, and the verification lift’s architecture is an open problem, not a given.

Raw material. This chapter’s existing text; MATH.TOOLBOX.md I.8 (the trichotomy and the promotion gate), I.5 (the derivation of $\gamma(\emptyset)$ for unchecked ticks); Chapter 20 of this book, which now owns the open question.

Steps.

1. **Retitle.** “The External Oracle” presumes V0’s architecture, where verification was one organ among several and could be an outside service. Rename to something that does not assume where the checker lives.
2. **Keep the trichotomy and its argument.** Two outcomes conflate REFUTED with UNVERIFIABLE — “the error of concluding falsehood from inability to check” — and would spike conflict over the checker’s own blind spots. Three outcomes fix it: the obstruction clears, increments, or is left untouched. A checker crash returns UNVERIFIABLE, never a refutation. **[PROVED]** as an epistemic argument.
3. **Keep the consolidation gate.** $\gamma(\nu)$ reads the verdict and is what stops unverified structure from entering the invariant. This is the link that makes verification load-bearing rather than advisory — state it as such, cross-referencing Chapter 8.
4. **Keep the derivation for unchecked ticks.** “No test ran” is absence of evidence, not a failed test, and the rate for such ticks was *derived* by shrinkage rather than hand-set, degrading exactly to the old constant on day one. Keep the honest caveat: it is conditional on selection bias, since ticks that ran a check may not resemble ticks that did not. Testable, strictly better than a constant, never to be quoted as unbiased.
5. **State the implementation status.** The trichotomy is computed; it has never been connected to a real prover or sandbox, so the verdict is not earned from evidence in any live loop. **[PROPOSED]**.
6. **Hand the open question to Chapter 20** rather than answering it here, and say so explicitly so the two chapters do not drift apart.

Done when. The trichotomy is stated with its epistemic argument, its link to the invariant is explicit, and no text presumes an external service.

Intuition. Coherence governs whether to keep exploring or stop and reconcile. Only an

external checker touches *truth*. And “I could not check” must never be mistaken for “it is false”.

Definition 17.1 (Three-valued verification). `VERIFYOP` calls a sandboxed symbolic oracle returning one of three verdicts, acting on an obstruction counter Ω :

exit	verdict	effect on state
0	VERIFIED	obstruction cleared; state may crystallise
1	REFUTED	$\Omega += 1000$; forced into <code>RESOLVE</code>
2	UNVERIFIABLE	Ω <i>untouched</i>

Remark 17.2 (Absence of proof is not proof of absence). An earlier version treated every nonzero exit as failure, conflating `REFUTED` with `UNVERIFIABLE`. That would spike conflict over the oracle’s own blind spots and lock the system into `RESOLVE` over perfectly good mathematics. A checker crash also returns `UNVERIFIABLE`. This three-valued logic is the promotion gate of Construction 10.6: a composite becomes a generator of $\mathfrak{D}$ only if it survives the oracle, and surviving means `VERIFIED`, not merely “not refuted”.

Claims and non-claims. The asymmetry in the table is deliberate and is the load-bearing design decision. A refutation moves Ω by a large fixed amount; an inability to check moves it not at all. The system is therefore allowed to be ignorant without being punished for it, and is punished immediately for being wrong. The alternative — treating unverifiable as mildly bad — produces a gradient toward only ever asserting things the oracle happens to understand, which is a narrower system, not a better one.

Chapter 18

The Controller

Definition 18.1 (Coherence-gated mode). With threshold ε , the controller selects

$$\text{mode}(x) = \begin{cases} \text{RESOLVE}, & \rho > \varepsilon, \\ \text{EXPLORE}, & \rho \leq \varepsilon, \\ \text{UNKNOWN}, & \rho = \perp. \end{cases} \quad (18.1)$$

Remark 18.2 (Three modes, and an empirical ε). The UNKNOWN branch is essential: when ρ is undefined the controller *refuses* to default to EXPLORE, because “no measurement” is not “all is well” (Remark 5.5). The threshold is calibrated rather than guessed: on real 384-dimensional embeddings, agreement gave $\rho \approx 0.00$, related-but-distinct ≈ 0.06 , and off-topic ≈ 0.14 , so $\varepsilon \approx 0.10$ is a principled starting gate.

Two cautions were recorded from live runs. First, ρ is *not* monotonic in the number of active organs: adding a well-aligned organ raises the normaliser faster than the discord, so ρ dilutes. Second, comparisons across differently sized complexes must be made with care, because B and $\|X\|_F^2$ both change.

Remark 18.3 (Superseded: gate on $\tilde{\rho}$, not ρ). Both cautions are consequences of a single fact, isolated in Proposition 5.6: $\rho = \alpha \cdot \tilde{\rho}$, and *only α has the pathologies*. Organ-count dilution moves α and leaves $\tilde{\rho}$ invariant. Cross-complex comparison is unsafe for ρ , but $\tilde{\rho}$ is comparable across complexes once expressed as its position inside the window (5.4). The controller should therefore gate on $\tilde{\rho}$ — or on its window position — and report α alongside as the answer to a genuinely different question.

The typed reading is what the single number could not give:

- large α , small $\tilde{\rho}$: a clean two-camp split; the coalition wants restructuring (**split**, or a new **bind**);
- small α , large $\tilde{\rho}$: one organ is an outlier; a single edge wants repair.

Worked example. Apply the controller to the running example. With $\varepsilon = 0.10$ and $\rho = 0.2$, mode (18.1) returns RESOLVE, and a diagnostic reading only ρ would report “mild disagreement, reconcile”. The typed reading gives more: $\alpha = 0.5$ is large and $\tilde{\rho}$ sits at window position $\frac{1}{3}$, so by Remark 18.3 this is a two-camp split, and the correct remediation is **split** or a new **bind** — *not* repair of the worst edge. Note also that the worst-edge diagnostic was ambiguous here (the argmax tied), which is the same fact appearing in a

second instrument. □

Claims and non-claims. The controller is the one place where every earlier honesty decision becomes operational. $\perp$ propagates into a mode rather than being coerced to a number; the split of Proposition 5.6 changes which quantity is gated on; the window of Proposition 5.8 makes the threshold topology-relative. It is also where a mistake is most expensive, because a controller that mis-reads its instrument does not merely report wrongly — it acts.

Part XI

Reproduction

Chapter 19

Reproduction

NEW IN V2 *The weakest chapter in the book, placed last and labelled as such. It proposes a canonical morphism and claims nothing beyond that.*

Work plan. NEW IN V2

Goal. Give “the kernel reproduces itself” a precise candidate definition, and be explicit that a canonical morphism is not yet a viable offspring.

Raw material. TATIANA.V1.IDENTITY.MAP.md §5 (the proposal and its own warning), §10 open decision 4; MATH.TOOLBOX.md II.2 (the adjunction, which is standard and already proved).

Blocked by. Chapter 8 (there is no sub-invariant to carry until $K(\mathbb{K})$ is established) and Chapter 9 (there is nothing for an offspring to *do* until the wake/sleep cycle is specified). Do not write this chapter before those two are finished.

Steps.

1. **Say what reproduction is not.** Not a copy: a bitwise duplicate carries no information about what the kernel learned to be, and duplicating the whole store is not growth but storage. State this before proposing anything.
2. **Propose the morphism.** The adjoint pair is already in the mathematics: ι^* instantiates a downward-closed subcomplex, $\iota_!$ writes back only what was touched, and because ι is a full subposet inclusion the unit is an isomorphism, $\iota^* \iota_! \cong \text{Id}$ — so the round trip provably does not corrupt the fragment. An offspring is a downward-closed subcomplex together with its pullback sheaf.
3. **State the invariant it carries.** The offspring’s K is the restriction of the parent’s dimension vector to the subcomplex — a *sub-invariant*. Establish that this is well-defined and that it is conserved by the offspring’s own consolidation, which follows from Chapter 8 applied to the subcomplex.
4. **Do not claim viability.** Tag the whole chapter **[PROPOSED]**. The honest statement is: the morphism is canonical, and nothing follows from that about whether the offspring can run, learn, or survive contact with input. Say so in those words.
5. **Write the falsifier.** Before any implementation: what observation would show an offspring is *not* viable? Candidates to develop — the offspring’s wake phase never stalls (so it can never grow); or it stalls immediately everywhere (so its address carries no information); or its invariant drifts where the parent’s did not. Specify at least one as a runnable experiment.
6. **Connect to the selection question, and leave it open.** If reproduction is cheap and viability is rare, something must choose which fragments reproduce. Note that V0 explicitly **[REFUTED]** random variation-and-selection as the growth mechanism — variation needs randomness *because it has no address*, and this architecture computes an address — so the same argument constrains

what a selection rule here may look like. Do not propose one yet.

Done when. The morphism is defined, the sub-invariant is proved to be well-defined, the viability claim is explicitly *not* made, and at least one falsifier is specified precisely enough to run.

Part XII

The Encoder and the Verification Lift

Chapter 20

The Encoder and the Verification Lift

NEW IN V2 *Two roles the architecture cannot do without and has not yet derived. This chapter exists to keep them visible as open problems rather than letting them be quietly filled by an import.*

Work plan. NEW IN V2

Goal. State precisely what the encoder and the verification lift must *do* — as constraints the rest of the mathematics imposes on them — without assuming any particular architecture for either.

Raw material. TATIANA_V1_IDENTITY_MAP.md §2.1 (the narrow allowance, now further narrowed — see §1.5 of Chapter 1); MATH_TOOLBOX.md I.8 (the verification trichotomy, already fixed epistemically), I.5 (what the drive term needs to be for the wake flow to act on it).

Steps.

1. **Derive the encoder's type, do not choose it.** Work backwards from the wake phase: the drive term enters $\dot{s} = -L_{\mathcal{F}}s + \text{drive}$, so o_t must live where the restriction maps can receive it. Establish the minimum: which space, what normalisation, and whether anything more than a vector is required. The answer is a *constraint*, and the chapter's contribution is to state it exactly.
2. **Ask what would make an encoder wrong.** Two failure modes to develop: an encoder whose output geometry already contains the structure the sheaf is supposed to learn (in which case the sheaf is decorative and the architecture is a dressed-up embedding lookup), and an encoder whose output is isotropic noise to the sheaf (in which case nothing can ever glue). Both are checkable. Specify how.
3. **Derive the verification lift's contract.** What must $\mathcal{L}$ guarantee for the trichotomy — VERIFIED, REFUTED, UNVERIFIABLE — to remain meaningful? The third value is the load-bearing one: it exists precisely so that inability to check is never recorded as falsehood, and a checker crash must return it rather than a refutation. State the contract as requirements on the lift, not as a description of a particular checker.
4. **Connect the contract to consolidation.** $\gamma(\nu)$ reads the verdict: γ_0 if verified, $\varepsilon\gamma_0$ if unverifiable, 0 if refuted. So the lift's guarantees are exactly what stops unverified structure from entering the invariant. Make the dependency explicit — it is why this chapter cannot be deferred indefinitely.
5. **Check the surviving toolbox before reaching outward.** Ask whether either role reduces to something already present: is an encoder a restriction map into the complex from a boundary cell? Is a checker a discrete verification map with the same signature as the operators the book already defines? Record the answers even if negative, since a negative answer is what would

justify building something new.

6. **State the standing constraint, in the chapter, not in a footnote.** Neither role may be discharged by wrapping a pretrained, externally-hosted model. If a derivation arrives at machinery that also appears in such models, that is permitted — as a conclusion with a derivation attached, never as a default and never as an import.

Done when. Both roles have stated contracts derived from what the rest of the architecture requires of them; at least two failure modes are specified as checkable; and no architecture has been assumed for either.

Part XIII

Limits: What Is and Is Not Claimed

Chapter 21

Limits: What Is and Is Not Claimed

Work plan. ¡RECAST!

Goal. Keep this chapter’s function — the honest boundary — and update its content to V2’s boundary, which is in some ways wider and in most ways narrower than V0’s.

Raw material. This chapter’s existing text; MATH.TOOLBOX.md Part VIII (the consolidated dead-end registry), Part VI (four failed cover-detection attempts), Part IX (the methodology findings); TATIANA_V1_IDENTITY_MAP.md §2, §11, §12.

Steps.

1. **Replace the scope statement.** The V0 text scopes to three papers and a research-tool track. That track is retired. State V2’s scope: one architecture, evaluated by whether its invariant survives its own dynamics.
2. **Import the dead-end registry wholesale** into Appendix C and reference it here. Everything refuted, rejected or dissolved, with the reason, so none of it is re-proposed without knowing why it died. This is one of the project’s genuine assets.
3. **Keep the cover-vs-partition saga at full length.** Four instruments, all inconclusive, each diagnosed: an aggregation unit that fixed the answer before any data was seen; a null model with free row margins that manufactured a false positive on identical data ($p = 0.45$ versus $p = 0.02$ purely by changing the null); an instrument blind without dimensionality reduction; a label/score relation mismatch. State the acceptance criterion earned from it: any instrument must pass a synthetic positive, a synthetic negative *with planted structure recovered in the right direction*, and a structure-free geometric control, at one shared parameter setting, *before touching real data*. None of the four met it.
4. **Keep “four failures to reject the partition are not evidence that the partition is true.”** Verbatim.
5. **State the live external threat and the defence.** Published work reports that identity maps can replace learned sheaf Laplacians on a benchmark family. It concerns *graphs*. The 2-cell argument of Chapter 2 is the defence and must appear before a reader raises the objection — and it must be stated as a theorem, since as of now it is asserted.
6. **Reproduce the kill-tests as a live checklist**, with current status: routing (unrun), localisation on real data (unrun), invariant drift ([MEASURED], survived, with pruning untested), sheaf-not-decorative (unrun). Anything that fires goes in the logbook and in Appendix C.
7. **Keep the prior-art honesty.** Cover detection has mature machinery in at least four fields under different names, and roughly a week was spent rediscovering it. What remains genuinely original is the sheaf over the structure, the growth law, the accumulating-not-recomputing memory, and the two-complex architecture. Infrastructure should be borrowed, not invented.
8. **Carry the citation warning.** Nearly every citation in the source corpus is flagged as un-

searched. Verify before any bibliography.

Done when. A reader finishing this chapter can state exactly what has been established, what has been refuted, and what has merely not yet failed.

Intuition. The deepest thing the topology buys is not that it makes the system smart. It is that it makes the claim of augmentation *falsifiable*: memory, consistency and growth become operations on one object, with cheap provable guarantees and a live measure of whether augmentation is actually happening. A framework that cannot fail is not doing any work, and most of this chapter is about where this one can.

21.1 The bet and the refusal

Claims and non-claims. The bet (defensible). A persistent, self-checking, self-restructuring *organiser* wrapped around a fixed language model can, over time, produce better-grounded and more internally consistent research artifacts than the raw model, through measured cross-organ error correction and accumulation. The advantage is *architectural*, expressed as operations on the single object $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$.

The refusal. No simplex proves a theorem and no Laplacian integrates a differential form. The per-inference reasoning quality is bounded by the underlying model. The mathematics here does not raise that ceiling; it organises, measures and error-corrects *around* it.

The bet is about the *slope*, not the intercept. That makes it an empirical claim, and Open Problem 22.3 records that it has not been tested.

21.2 The scope is three papers, not one

Claims and non-claims. A single document containing the ρ splitting, the operator algebra, the coned concept, the cohomological obstruction and a spectral prediction on $\mathbb{CP}^3$ will produce something no referee can accept, because those claims have incompatible standards of evidence. The audit's recommended split is:

Paper A — a cellular-sheaf coherence measure for multi-agent orchestration. Content: $\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$; ω and ρ ; the $\alpha/\tilde{\rho}$ splitting with the window; weighted Anderson–Morley *with* the $\|\mathcal{F}\| \leq 1$ hypothesis stated; worst-edge localisation; the three-valued verification gate. Every claim is a theorem plus a measurement on a running system. This paper is nearly writable today.

Paper B — sheaf-cohomological obstruction as a diagnostic. Content: Proposition 7.3 — the derived cochain is exact, a negative result worth publishing on its own; the architecture needed to produce non-derived 1-cochains; the Hodge trichotomy; and the connection to contextuality in the Abramsky–Brandenburger sense, which is the thread that would make this a quantum-information contribution rather than a software paper. This paper *requires new engineering* and cannot be written before that lands.

Paper C — the spectral work. This is separate mathematics. It has nothing to do with TATIANA and should not be contaminated by it; the architecture's role is to be the

tool that helped, mentioned in a methods note, not a co-author of the theorem.

21.3 What must not be claimed

Carrying forward the honesty discipline that is the best thing about the source corpus:

- **Not** “we prove the whole exceeds the sum”. A structural novelty statement says a composite is *new*, not that it is true or useful. Verification remains the sole arbiter and runs first.
- **Not** “capacity measures learning”. It is gameable by adding independent useless operators. Report it; never optimise it.
- **Not** “the system uses D -module theory”. It uses Gaussians, which *are* the simplest holonomic modules — an exact identification that currently buys three invariants, all of them constant. The defensible claim is that the concept type admits a conservative extension to holonomic modules, feasible only because of the existing low-rank design.
- **Not** “Riemann–Hilbert applies”. One leg is rigorous (cellular sheaves are constructible sheaves via exit paths); the other requires a commitment to an algebraic stratification that has not been made.
- **Not** “the threshold is derived”. It is *normalised* (Proposition 5.8).
- **Not** “ b_1 measures conceptual gaps” without Remark 7.2. The justification is the whole content of the claim.
- **Not** “the architecture is an operad”. See Remark 12.2 and the three defects following it.

21.4 The findings, restated as boundaries

The audit’s fourteen findings are not a list of bugs; each marks a place where a claim outruns its evidence. Restated as standing limits:

1. **The audit itself was against a stale snapshot.** Several findings were repaired before the audit was read. Any use of this list must be re-checked against the current branch — and that instruction applies to this book.
2. **Theorem 7.5 is stated more generally than proved.** The gap between hypothesis and proof has not been located precisely, and until it is, the theorem should be cited in its proved form only.
3. **The $H^1(\mathcal{C}; \mathcal{F})$ repair is half circular.** The circularity must be identified before the repair is adopted (Open Problem 22.2).
4. **Proposition 12.4 is false in the deployed case.** A counterexample exists and must be reproduced in Appendix C before that entry is complete.
5. **The K_0 classes are trivial for every object the system produces.** An invariant that separates nothing is not a diagnostic (Open Problem 22.7).
6. **The independence relation is not symmetric,** its supports are placeholders, and its empty-set convention is inverted (Chapter 12).
7. **The restriction maps are the identity,** so the sheaf is a graph Laplacian in disguise (Open Problem 22.1).

Claims and non-claims. The honest summary of the whole corpus: the *measurement* layer is real, proved and running — ω , ρ , the splitting, the window, the three-valued gate, the coned concept and the growth law all survive scrutiny. The *expressive* layer is not yet built: while restrictions are the identity and 1-cochains are derived, the sheaf and its cohomology are carrying no information the graph did not already carry. That is not a fatal criticism; it is a precise statement of what the next piece of work has to be, and the fact that it can be stated this precisely is what the formalism bought.

Part XIV

Open Problems

Chapter 22

Open Problems within MOS

Work plan. **RECAST**

Goal. Re-rank the open problems for V2 and add the ones the pivot created. Retitle: “within MOS” names the retired system.

Raw material. This chapter’s existing text; TATIANA_V1_IDENTITY_MAP.md §10 (the ranked open decisions); MATH_TOOLBOX.md II.1 (Tier C material and why it was inert), III.5 (the contextuality bridge); TATIANA-V2/MATH/phase0_invariant_dimension.md §4.

Steps.

1. **Rank, and say what the ranking means:** an open problem here is one where a construction or a measurement is missing, not merely one where work remains.
2. **#1 — the source of a non-exact η for a single kernel.** V0 got its measured 1-cochain from organs disagreeing. V2 has no organs. Without a source, Proposition 8.2 leaves growth with no address and the kernel cannot grow at all. This is the most important open problem in the book. Candidate to develop and test: disagreement between the kernel’s own competing candidate transformations.
3. **#2 — co-allocation versus separation.** When a harmonic residual names a cycle, does the kernel cone it (integrate) or keep the memories distinct? The allocation literature says excitability history decides, which in this formalism means it depends on the coupling weights — and those weights are *computed and never read*. That is a live bug, not merely a gap.
4. **#3 — can molding erode the obstruction growth depends on?** Raised by the Phase 0 experiment, which bounds molding to one nudge per tick and so never tests it. If slow molding can dissolve a harmonic class, the layer handoff is a race rather than a handoff. State the experiment that settles it.
5. **#4 — does Ext^2 vanish for the incidence algebra of the 2-complex?** This decides whether “triangles make molding obstructible” is a theorem or an analogy. Quiver categories are hereditary; poset categories with relations should have higher global dimension. Currently unverified — do not build on it either way.
6. **#5 — the MDL exchange rate,** and **#6 — the stall tolerance:** the design’s only two admitted free parameters. Keep them listed so that a third would be conspicuous.
7. **#7 — reproduction viability** (Chapter 19), stated as currently carrying no claim.
8. **Revisit the shelved homological material on purpose.** Characteristic cycles, b -functions and Hochschild cohomology were shelved as inert for one stated reason: every concept the V0 engine produced was simple, so the machinery had nothing to act on. A coned concept is a limit of a diagram and is not obviously simple, so the premise may no longer hold. Re-examine, carrying the standing corrections so they are not re-derived wrongly — extension obstructions live in HH^3 and not HH^2 ; Ore extensions are unobstructed given their data; and a classification

theorem used here needs an algebraically closed field and was once silently applied over $\mathbb{R}$.

9. **Keep the contextuality bridge as a well-posed problem.** Both halves are standard separately; the intersection is not an established area. State the real gap sharply: a sheaf of vector spaces always admits the zero section, so this obstruction is about *dimension*, while the contextuality one is about *existence*.

Done when. The problems are ranked by what blocks what, and #1 is unmistakably identified as the one that gates the architecture.

These are stated as problems, not as directions, because each one has a definite answer that is currently unknown and whose resolution would change what the system can claim. They are ordered by how much else depends on them.

Open Problem 22.1 (Non-identity restriction maps). Find a procedure that learns restriction maps $\mathcal{F}_{v \leq e}$ from organ data such that (i) the maps are not all the identity, (ii) functoriality (3.1) holds on every filled simplex, and (iii) the learning is closed-form or otherwise costs no gradient training. Closed-form Procrustes alignment satisfies (i) and (iii) but is not known to satisfy (ii). Everything expressive about the sheaf is downstream of this problem: while the maps are the identity, $\ker L$ is the diagonal and the sheaf is a graph Laplacian in disguise (Proposition 4.6).

Open Problem 22.2 (A non-vacuous $H^1(\mathcal{C}; \mathcal{F})$). Under identity restrictions the cohomological hierarchy carries no information: the 1-cochain the engine emits is δx for a global state x , and a coboundary is exact by definition, so its class in $H^1(\mathcal{C}; \mathcal{F})$ vanishes identically — for every state, every sheaf, and every choice of restriction map. Find a source of 1-cochains that is *not* of the form δx — pairwise judgements formed independently of any global state are the candidate — and determine whether the resulting classes are stable enough to serve as a diagnostic. The audit records that one proposed repair is half circular; the circularity must be identified precisely before a fix is adopted.

Open Problem 22.3 (Does the architecture actually compound?). The central bet is on the *slope*, not the intercept: that a self-checking, self-restructuring organiser improves with use even though its per-inference ceiling is fixed by the underlying language model. This is an empirical claim and it has not been tested. Design an experiment with a pre-registered metric, a control (the same model with no architecture), and a horizon long enough for accumulation to show, and state in advance what result would falsify the bet.

Open Problem 22.4 (The right normaliser). The Anderson–Morley bound B over-bounds $\lambda_{\max}$, which compresses ρ toward zero and contributes to its dead dynamic range. Determine whether a tighter, still eigensolver-free bound exists for the sheaf Laplacian with non-identity restrictions, or prove that no degree-based bound can be tight in this regime.

Open Problem 22.5 (Stability of the diagnostics under growth). The growth operators change the topology, and both the calibration window and the kernel projection are topology-dependent. Give a stability estimate: if *bind* or *collapse* changes the complex by one simplex, how much can $\tilde{\rho}$ move for a fixed state? Without such an estimate, a change in $\tilde{\rho}$ across a growth step cannot be attributed to the state rather than to the rewiring.

Open Problem 22.6 (Functoriality repair by hybridisation). When restriction maps are obtained as projections onto an interface, functoriality fails predictably. The canonical repair is to promote the interface to a cell of its own, which is what hybridisation methods do in the finite-element setting. Determine whether the same repair applies to the cognitive complex, and what it costs: does promoting every offending interface preserve the two-level stratification, or does it force a third stratum?

Open Problem 22.7 (Composition factors as memory atoms). A later development formalises memory atoms as Jordan–Hölder composition factors, with retrieval ordered by stratum, then integer content, then structure. Determine whether the composition series is canonical for the objects the system actually produces — the audit records that the associated K_0 classes are trivial for every such object, which if correct means the invariant distinguishes nothing.

Part XV

Potentials for Quantum Implementation

Chapter 23

Potentials for Quantum Implementation

Work plan. **¡KEEP!**

Goal. Keep as a forward-looking chapter, unchanged in substance. Guard it against the one failure mode it invites: becoming a claim rather than a programme.

Raw material. This chapter's existing text; `MATH_TOOLBOX.md` I.4 (restriction maps as CPTP channels — the structural reason this chapter is not fanciful), III.5 (the proposed toy object).

Steps.

1. **Keep the structural motivation, which is genuine.** Storing a stalk as a rank- k SPD matrix makes it the cone over density-matrix space; congruence by an orthogonal map preserves trace, so restriction maps already *are* unitary channels. The quantum reading is not imported — it is what the objects already are. **[PROVED]**.
2. **Keep the obstruction section prominent.** Input and output remain the hard part; a block-encoding that cannot be loaded or read is not an implementation. This section is the reason the chapter is honest and it must not shrink.
3. **Keep the toy object as a proposal.** A cellular sheaf with quantum-state stalks, whose Laplacian is a Hamiltonian, whose ground space is the globally consistent assignments and whose ground-state energy is a contextuality witness. **[PROPOSED]**, clearly.
4. **Add a status line at the head of the chapter.** Nothing here is on the critical path for V2. It is a direction, retained because the architecture points at it for structural reasons, not because it is being built. Say so explicitly so that a reader in a hurry cannot mistake it for work in progress.

Done when. The chapter's structural motivation is stated as proved, everything else as proposed, and its status relative to V2's actual work is unambiguous.

Claims and non-claims. **[PROPOSED]** Nothing in this part is implemented, and nothing in it is required by any result elsewhere in the book. It is included because the sheaf Laplacian is exactly the kind of object for which quantum linear algebra has a story, and because that story has a specific shape worth recording before it is attempted. Every claim here should be read as a research programme with an identified failure mode, not as a capability.

23.1 Why this object at all

The two quantities TATIANA computes on every tick are a Dirichlet energy $\omega(x) = x^\top Lx$ and a normalised version of it. Both are quadratic forms in a sparse, symmetric, positive semidefinite operator built from local data. This is the standard input to quantum linear-algebra routines, and the standard caution applies: the speedups are in the linear algebra, and they are destroyed by the cost of getting classical data in and scalar answers out.

23.2 The Dirac operator and its block-encoding

Definition 23.1 (Sheaf Dirac operator). Let $d_{\mathcal{F}} = \bigoplus_k \delta^k$ be the total coboundary of the sheaf. The *sheaf Dirac operator* is

$$D_{\mathcal{F}} = d_{\mathcal{F}} + d_{\mathcal{F}}^\dagger,$$

a self-adjoint operator on $\bigoplus_k C^k(\mathcal{C}; \mathcal{F})$ satisfying $D_{\mathcal{F}}^2 = \Delta$, the total Hodge Laplacian. Its kernel is the space of harmonic cochains, which by Hodge theory is isomorphic to $\bigoplus_k H^k(\mathcal{C}; \mathcal{F})$.

The programme, stated as a pipeline:

1. **Block-encode** $D_{\mathcal{F}}$. Construct a unitary U whose top-left block is $D_{\mathcal{F}}/\gamma$ for a subnormalisation $\gamma \geq \|D_{\mathcal{F}}\|$. For a cellular sheaf with *matrix-valued* restriction maps this is the substantive step, and the subnormalisation γ scales with the heterogeneity of those matrices — which is precisely the quantity that makes the sheaf interesting in the first place.
2. **Filter onto** $\ker D_{\mathcal{F}}$. Apply a polynomial approximation to a projector via quantum singular value transformation, isolating the harmonic subspace.
3. **Extract a scalar**. Estimate ω , or an overlap with a prepared state, by amplitude estimation.

Remark 23.2 (Where this programme is honest and where it is not). Step 1 is the only step with genuine novelty for cellular sheaves with matrix-valued restrictions; steps 2 and 3 are standard. The honest statement of the advantage is therefore about the *encoding* and its resource scaling, not about an end-to-end speedup for TATIANA. In the deployed regime the restriction maps are the identity (Proposition 4.6), $L = L_K \otimes I_d$, and the whole object reduces to a graph Laplacian on seven vertices — for which classical computation is instantaneous and any quantum treatment is theatre. The quantum question only becomes non-trivial once Open Problem 22.1 is solved and the maps are genuinely matrix-valued and heterogeneous.

23.3 The obstruction: input and output

Claims and non-claims. Two costs dominate and both are classical.

Input. The stalk data is a set of 384-dimensional embeddings produced by a classical encoder. Preparing an amplitude encoding of these costs, in the absence of a QRAM, time linear in the data — which is the same order as computing ω classically. Any claimed speedup that assumes free state preparation is assuming away the problem.

Output. ω is one real number, and amplitude estimation returns it to precision ε in $O(1/\varepsilon)$ queries. For a diagnostic thermometer read once per tick, ε need not be small, but

neither is the classical cost large. The regime where the quantum route can win is not “compute ρ faster”; it is “compute something about the spectrum of L that is classically infeasible”, and no such quantity is currently required by the architecture.

23.4 What would make this real

Open Problem 23.3 (A quantity worth the encoding). Identify a functional of L that (i) the architecture genuinely needs, (ii) is classically infeasible at the scale TATIANA reaches, and (iii) is extractable from the block-encoded $D_{\mathcal{F}}$ by a constant number of scalar measurements. Without (i) the programme is solving an unposed problem; without (ii) it is slower than the classical route; without (iii) the output cost swamps the encoding.

Open Problem 23.4 (Subnormalisation in terms of sheaf data). Give an explicit bound on the block-encoding subnormalisation γ for a cellular sheaf Dirac operator in terms of the restriction-map norms and the degrees of the complex, and determine how it degrades as the maps become heterogeneous. This is the resource-scaling statement on which any advantage claim rests.

Remark 23.5 (Relation to the eddy-current programme). The block-encoding of a cellular sheaf Dirac operator with matrix-valued restriction maps is the technical core of a separate research programme on magnetoquasistatic problems, where the sheaf arises from a discretised PDE rather than from a cognitive architecture and the subnormalisation scales in material contrast. The mathematics is the same; the physical problem supplies what Open Problem 23.3 asks for and TATIANA, at present, does not. That asymmetry is the honest reason this part is short.

Appendices

Appendix A

Notation

Symbol	Meaning
$\mathcal{M} = (\mathcal{C}, \mathcal{F}, s)$	cognitive state space
$\mathcal{C}$	stratified base complex
K	coarse complex (organs as vertices)
$\mathcal{C}_v$	fine complex inside organ v
$\sigma \trianglelefteq \tau$	σ is a face of τ
$\mathcal{F}$	cellular sheaf on $\mathcal{C}$
$\mathcal{F}(\sigma)$	stalk over σ
$\mathcal{F}_{\sigma \trianglelefteq \tau}$	restriction map $\mathcal{F}(\sigma) \rightarrow \mathcal{F}(\tau)$
$C^k(\mathcal{C}; \mathcal{F})$	k -cochains $\bigoplus_{\sigma \in \mathcal{C}_k} \mathcal{F}(\sigma)$
δ	coboundary $C^0(\mathcal{C}; \mathcal{F}) \rightarrow C^1(\mathcal{C}; \mathcal{F})$
$L = \delta^\top \delta$	degree-zero sheaf Laplacian
$\omega(x) = x^\top Lx$	discord (Dirichlet energy)
ρ	coherence score in $[0, 1]$
α	dissent fraction
$\tilde{\rho}$	mode index
B	Anderson–Morley normaliser
π_e	edge precision
$\perp$	undefined / measurement not made
b_0, b_1	Betti numbers of the 1-skeleton
$H^k(\mathcal{C}; \mathcal{F})$	k -th sheaf cohomology
$\mathfrak{D}$	operator algebra of growth operators
$D_{\mathcal{F}}$	sheaf Dirac operator $d_{\mathcal{F}} + d_{\mathcal{F}}^\dagger$
d	stalk dimension (384 in deployment)

Appendix B

Discrepancy Register

Places where the sources disagree with each other, or where the implementation determines a mathematical fact that the prose states differently. Each is a fact about TATIANA, not a typo.

D1 — Concepts “inside the stalk”. `TATIANA_LOGBOOK.md` §0.1 describes the fine level as “concepts inside each organ’s stalk”. The stalk is a vector space; the fine complex is a simplicial complex. Conflating them makes the coarse-graining π_v look like an identity when it is a lossy precision-weighted fusion. Resolved in favour of the separation (Remark 3.7).

D2 — Restriction maps point the wrong way in the poster. `POSTER/sections/05_sheaf.tex` defines $\rho_{\tau \rightarrow \sigma} : \mathcal{F}(\tau) \rightarrow \mathcal{F}(\sigma)$ and draws the arrows downward, then asserts $s \in H^0(\mathcal{C}; \mathcal{F})$. These are incompatible (Proposition 3.1); the downward convention defines a cosheaf. `TATIANA_Mathematical_Description.tex` has it correct. **Action: fix the poster before showing it.**

D3 — Two different systems named MOS. The `README` on `main` describes a retrieval pipeline (EvidenceEngine, KnowledgeCurator, Canonicalizer). The poster and the mathematical description define a sheaf-theoretic state space. These are different objects sharing a name, and the `README` is what a reader sees first.

D4 — `main` is stale. At the time of writing, `main` is four commits and roughly six weeks behind `fix-11-13-and-curvature-decisions`, which carries +33,450 lines including all of Part VI’s material and the `PRECILLA` package. Any reader who clones the default branch gets a snapshot that this book does not describe.

D5 — `hodge.py` is scalar, not sheaf-valued. The Hodge decomposition code operates on a scalar complex — the trivial sheaf $\mathcal{F}(\sigma) = \mathbb{R}$ — with $(\delta^0 f)_{uv} = f(v) - f(u)$. It is therefore a computation about the *shape*, not about the sheaf. This is correct for what it measures and should not be cited as evidence about sheaf-valued cohomology.

D6 — The Hebbian-weight gap appears to be closed. An earlier scrutiny session recorded that plasticity weights were computed but never read by the coherence measurement, which would make ρ a different functional from the one documented. The current `coherence.py` computes a precision-weighted degree $d_\pi(v) = \sum_{e \ni v} \pi_e$ and a weighted Anderson–Morley bound,

so the gap looks repaired. **This needs confirming against the precision source** before the finding is retired.

Appendix C

Refuted-Claims Ledger

Claims that were made, tested, and failed. They are kept because the refutations are the expensive part.

R1 — Encoder anisotropy explains the dead dynamic range of ρ . [REFUTED] (2026-07-28). The deployed encoder returns unit-normalised vectors, for which $\alpha = (n - 1)(1 - \bar{c})/n$ exactly, $\bar{c}$ the mean pairwise cosine. At the observed $\bar{c} \approx 0.449$ with $n = 7$ this gives $\alpha \approx 0.46$, which is not small. The compression is produced jointly by α , by $\tilde{\rho}$ sitting mid-window, and by B over-bounding $\lambda_{\max}$.

R2 — The cohomological hierarchy is informative as built. [REFUTED] The 1-cochain the engine emits is δx for a global state x ; a coboundary is exact, so its class in $H^1(\mathcal{C}; \mathcal{F})$ vanishes identically, for every state and every sheaf. The growth mechanism has no address in $H^1(\mathcal{C}; \mathcal{F})$. See Open Problem [22.2](#).

R3 — Proposition 12.4 of the earlier description. [CONTESTED BY AUDIT] The audit records it as false in the deployed case. The statement and the counterexample must be reproduced here before this ledger entry is complete.

R4 — Theorem 7.5 as stated. [CONTESTED BY AUDIT] The audit records that it is stated more generally than it is proved. The gap between hypothesis and proof must be located precisely.

R5 — K_0 classes as memory invariants. [CONTESTED BY AUDIT] The audit records that the K_0 classes are trivial for every object the system produces, which would mean the invariant separates nothing. See Open Problem [22.7](#).

Bibliography